

Supplementary Material: The Attribution Impossibility

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1 Complete Axiom System

Table 5 lists all axioms in the formalization (the core impossibility theorem requires none of them—only the Rashomon property as hypothesis). The formalization uses 6 axioms total: 2 type declarations (`Model`, `firstMover`), 3 domain-specific property axioms (`firstMover_surjective`, `crossGroupBaselineCore`, `proportionalityConstant`), and 1 measure-theoretic axiom (`modelMeasure`). Ten former axioms—including the split-count function, 4 split-count properties, attribution function, proportionality, boosting rounds, measurable space, and query-complexity constant—are now definitions or derived theorems (see axiom reduction history in main text).

2 Proof Architecture

The formalization comprises 58 Lean 4 files with 6 axioms and 357 type-checked theorems and lemmas (0 sorry). Of these, 139 require multi-line proofs (≥ 5 tactic lines); 6 are definitional wrappers or single-step applications. The core impossibility (`attribution_impossibility`) depends on zero axioms—only the Rashomon property as hypothesis. The DASH resolution (`consensus_equity`) depends on the `attribution_sum_symmetric` theorem (derived from axioms); the variance convergence depends on the `consensus_variance_bound` theorem (also now derived, not axiomatized).

```
Defs.lean (6 axioms, type definitions, derived theorems)
  +-- SymmetryDerive.lean (attribution_sum_symmetric, DERIVED)
  +-- Trilemma.lean (RashomonProperty, attribution_impossibility)
  |   +-- Iterative.lean, Lasso.lean, NeuralNet.lean
  |   +-- General.lean (GBDT impossibility)
  |   +-- UnfaithfulBound.lean ( $U \geq 1/2$ , ties optimal, path convergence)
  |   +-- RashomonUniversality.lean (Rashomon from symmetry)
  |   |   +-- RashomonInevitability.lean (inescapable)
  |   +-- AlphaFaithful.lean (alpha-faithfulness bound)
  |   |   +-- ConditionalImpossibility.lean (cond. SHAP)
  |   +-- FairnessAudit.lean (proxy audit = coin flip)
  |   +-- LocalGlobal.lean (local  $\geq$  global instability)
  +-- SplitGap.lean, Ratio.lean, SpearmanDef.lean
  +-- FlipRate.lean (exact GBDT flip rate, binary = coin flip)
  +-- Efficiency.lean (SHAP efficiency amplification)
  +-- Impossibility.lean, Corollary.lean, EnsembleBound.lean
  +-- DesignSpace.lean + DesignSpaceFull.lean (all 4 steps)
  +-- ModelSelection.lean + ModelSelectionDesignSpace.lean
  +-- SymmetricBayes.lean (GENERAL SBD, orbit bounds, trichotomy)
  +-- CausalDiscovery.lean (causal discovery impossibility)
  +-- SBDInstances.lean (abstract aggregation + instances)
  +-- FIMImpossibility.lean (Gaussian FIM  $\rightarrow$  Rashomon)
  +-- GaussianFlipRate.lean (Phi definition, flip rate formula)
```

Table 5: Complete axiom inventory (6 total).

Axiom	Lean 4 name	Justification
Model type	<code>Model</code>	Abstract type for trained models
First-mover function	<code>firstMover</code>	The dominant feature in each model
First-mover surjectivity	<code>firstMover_surjective</code>	DGP symmetry: each feature can be first-mover
Cross-group baseline	<code>crossGroupBaselineCore</code>	Split counts for features in other groups depend only on group indices
Proportionality constant	<code>proportionalityConstant</code>	$\varphi_j(f) = c \cdot n_j(f)$ with $c > 0$ uniform across models
Training measure	<code>modelMeasure</code>	Probability measure on Model (the training distribution)
<i>Formerly axiomatized, now defined or derived (10 eliminated):</i>		
Split count	<code>splitCount</code>	Definition: if-then-else on first-mover group membership
Split count (first-mover)	<code>splitCount_firstMover</code>	Theorem: derived from <code>splitCount</code> definition
Split count (non-first-mover)	<code>splitCount_nonFirstMover</code>	Theorem: derived from <code>splitCount</code> definition
Cross-group symmetry	<code>splitCount_crossGroup_symmetric</code>	Theorem: same <code>crossGroupBaselineCore</code> for same-group features
Cross-group stability	<code>splitCount_crossGroup_stable</code>	Theorem: <code>crossGroupBaselineCore</code> depends only on group indices
Attribution function	<code>attribution</code>	Definition: <code>proportionalityConstant · splitCount</code>
Global proportionality	<code>proportionality_global</code>	Theorem: derived from <code>attribution</code> definition
Boosting rounds	<code>numTrees</code> , <code>numTrees_pos</code>	Fields of <code>FeatureSpace</code> structure ($T, T > 0$)
Measurable space	<code>modelMeasurableSpace</code>	Instance: discrete σ -algebra ($\mathbb{T}$, works for any type)
Testing constant	<code>testing_constant</code> , <code>testing_constant_pos</code>	Definition: $C := 1/8$ from Tsybakov (2009); positivity by <code>norm_num</code>
<i>Additional derived theorems:</i>		
Spearman bounds	<code>spearman_instability_bound</code>	Theorem in <code>SpearmanDef.lean</code> (from midranks)
Attribution sum symmetry	<code>attribution_sum_symmetric</code>	Theorem in <code>SymmetryDerive.lean</code>
Variance bound	<code>consensus_variance_bound</code>	Theorem in <code>VarianceDerivation.lean</code>
Chebyshev query bound	<code>chebyshev_query_bound</code>	Theorem in <code>QueryComplexityDerived.lean</code> ($C = 1/12$)
Bayes-optimal tie	<code>bayes_optimal_tie</code>	Theorem in <code>BayesOptimalTie.lean</code>
Pareto optimality	<code>dash_pareto_optimal</code>	Theorem in <code>ParetoOptimality.lean</code>

```

+-- QueryComplexity.lean (query lower bound,
    Le Cam  $C \geq 1/8$ ; Chebyshev  $C = 1/12$  in Derived)
+-- QueryComplexityDerived.lean (Chebyshev alternative,  $C = 1/12$ )
+-- MeasureHypotheses.lean (independence + measure hypotheses)
+-- VarianceDerivation.lean (consensus variance from IndepEnsemble)
+-- UnfaithfulQuantitative.lean (unfaithfulness bounds)
+-- BinaryQuantizer.lean (binary quantization lemmas)
+-- MechInterp.lean (mechanistic interpretability impossibility)
+-- BeyondBinary.lean (extensions beyond binary explanation problems)
+-- BayesOptimalTie.lean (Bayes-optimal tie)
+-- LossPreservation.lean (loss preservation)
+-- ParetoOptimality.lean (DASH Pareto optimality)
+-- Consistency.lean (axiom system consistency, Fin 4 model)
+-- RandomForest.lean (contrast case, documentation only)
+-- Setup.lean (GBDTSetup, axiom-free API)
+-- Qualitative.lean (dominance + surjectivity)
+-- ProportionalityLocal.lean (per-model  $c$  only)
+-- ApproximateEquity.lean (Rashomon from  $K$ -ratio)
+-- QueryComplexityParametric.lean (parametric)
+-- RobustnessLipschitz.lean (Lipschitz in  $\rho$ )
+-- StumpProportionality.lean (depth-1, DERIVED)
+-- LocalSufficiency.lean (local  $c$  suffices)
+-- MutualInformation.lean (MI generalisation)
+-- IntersectionalFairness.lean ( $K$ -audit:  $1 - (1/2)^K$ )

```

2.1 Lean File Cross-Reference

Table 6 maps each Lean file to its corresponding supplement section and key theorem.

Design Space: now fully formalized. The Lean formalization now covers all 4 steps of the Design Space Theorem. `DesignSpaceFull.lean` proves exhaustiveness: every ranking is either Family $\mathcal{A}$ (complete + unfaithful) or Family $\mathcal{B}$ (tie). `strongly_faithful_impossible` formalizes the impossibility of consensus-value-faithful methods. The Bayes-optimality half (optimal unfaithful ranking assigns ties) is proved in the supplement below and is also formalized in `BayesOptimalTie.lean`. `strongly_faithful_impossible` formalizes the impossibility of consensus-value-faithful methods (a stronger result than the supplement’s Step 3, which uses ranking-faithfulness).

Variance: partially grounded in Mathlib. The single-model variance `attribution_variance` is now *defined* from Mathlib’s `ProbabilityTheory.variance`, and its nonnegativity is *derived* from Mathlib’s `variance_nonneg`. This required adding two infrastructure axioms (`modelMeasurableSpace`, `modelMeasure`) to connect the abstract `Model` type to Mathlib’s measure theory.

Table 6: Lean file cross-reference (58 files, 357 theorems, 6 axioms).

File	Supplement §	Key Result	Sorry?
Defs.lean	§1 (Axiom System)	6 axioms + derived theorems	No
Trilemma.lean	Theorem 1	attribution_impossibility	No
Iterative.lean	§3	iterative_impossibility	No
General.lean	§3	gbdt_impossibility	No
SplitGap.lean	Lemma 2	split_gap_exact	No
Ratio.lean	Theorem 3	ratio_tendsto_atTop	No
SpearmanDef.lean	§4	spearmanCorr_bound	No
Lasso.lean	§3.2	lasso_impossibility	No
NeuralNet.lean	§3.3	nn_impossibility	No
RandomForest.lean	§3.4	(documentation only)	No
Impossibility.lean	Combined	not_equitable, not_stable	No
Corollary.lean	Corollary 1	consensus_equity	No
SymmetryDerive.lean	Derived	attribution_sum_symmetric	No
DesignSpace.lean	Theorem 5	design_space_theorem	No
DesignSpaceFull.lean	Theorem 5 Step 3	family_a_or_family_b	No
ModelSelection.lean	§Model Selection	model_selection_impossibility	No
ModelSelectionDesignSpace.lean	§Model Selection	model_family_a_or_family_b	No
AlphaFaithful.lean	§Approx. bounds	alpha_faithful_bound	No
UnfaithfulBound.lean	§Unfaithfulness	stable_complete_unfaithful	No
PathConvergence.lean	§Path convergence	relaxation_paths_converge	No
RashomonUniversality.lean	§Universality	rashomon_from_symmetry	No
RashomonInevitability.lean	§Universality	rashomon_inevitability	No
ConditionalImpossibility.lean	§Conditional	conditional_attribution_impossibility	No
FlipRate.lean	§Exact flip rate	binary_group_flip_rate	No
Efficiency.lean	§Efficiency	amplification_bounds	No
LocalGlobal.lean	§Local vs. Global	local_attribution_impossibility	No
SymmetricBayes.lean	§SBD	symmetric_bayes_dichotomy	No
CausalDiscovery.lean	§Causal Discovery	causal_discovery_impossibility	No
SBDInstances.lean	§SBD Instances	abstract_aggregation_impossibility	No
FairnessAudit.lean	§Fairness Audit	fairness_audit_impossibility	No
EnsembleBound.lean	§F2 Pareto	variance_decreases_with_M	No
FIMImpossibility.lean	§F3 FIM	gaussian_rashomon_witnesses	No
GaussianFlipRate.lean	§F1 Diagnostic	Phi, gaussianFlipRate	No
QueryComplexity.lean	§Query complexity	query_complexity_lower_bound	No
QueryComplexityDerived.lean	§Query complexity	chebyshev_query_bound (C=1/12)	No
MeasureHypotheses.lean	§Variance	independence + measure hypotheses	No
VarianceDerivation.lean	§Variance	consensus_variance_derived	No
UnfaithfulQuantitative.lean	§Unfaithfulness	quantitative_unfaithful_bound	No
BinaryQuantizer.lean	§Binary	binary_quantizer_lemmas	No
MechInterp.lean	§Binary	mech_interp_bilemma	No
BeyondBinary.lean	§Binary	beyond_binary_extensions	No
BayesOptimalTie.lean	Theorem 5 Step 3	bayes_optimal_tie	No
LossPreservation.lean	§Optimality	loss_preservation_tie	No
ParetoOptimality.lean	§F2 Pareto	dash_pareto_optimal	No
Consistency.lean	§Axiom Consistency	axiom_system_consistent	No
Basic.lean	(import hub)	—	No
Setup.lean	§Axiom System	attribution_impossibility_bundled (GBDTSetup)	No
Qualitative.lean	Strengthening	impossibility_qualitative (2 axioms)	No
ProportionalityLocal.lean	Strengthening	gbdt_impossibility_local (per-model c)	No
ApproximateEquity.lean	Strengthening	approximate_equity (bounded K-ratio)	No
QueryComplexityParametric.lean	§Query complexity	query_complexity_parametric (no global axioms)	No
RobustnessLipschitz.lean	§Robustness	flip_rate_lipschitz	No
StumpProportionality.lean	§Proportionality	stump_proportionality_derived	No
LocalSufficiency.lean	§Axiom Stratification	local_c_suffices (quantitative bounds)	No
MutualInformation.lean	§Open Problems	impossibility_from_mutual_info	No
IntersectionalFairness.lean	§Fairness Audit	intersectional_compounding (K audits)	No

The consensus variance bound ($\text{Var}(\bar{\varphi}_j) = \text{Var}(\varphi_j)/M$) is now *derived* in `VarianceDerivation.lean` from an independence hypothesis on ensemble draws, rather than axiomatized. The derivation uses an `IndepEnsemble` hypothesis (stating that ensemble model attributions are independent and identically distributed) together with the textbook identity $\text{Var}(\bar{X}) = \text{Var}(X)/M$ for i.i.d. variables. The derived theorems (variance halving, monotonicity) are genuine Lean proofs from this hypothesis.

Spearman bound: three derived bounds. The Lean formalization now contains three Spearman bounds, all derived (none axiomatized): (i) `spearmanCorr_bound` (`SpearmanDef.lean`), fully derived from the definition of Spearman correlation, giving the original weaker bound $\rho_S \leq 1 - 3(m-1)^2/(P^3 - P)$; (ii) `spearmanCorr_bound_groupSize_tight` (`SpearmanDef.lean`), a tighter derived bound $\rho_S \leq 1 - 3m^2/(P^3 - P)$ obtained by tightening the group-size constant; and (iii) the classical bound $\rho_S \leq 1 - m^3/P^3$, which is *not yet formalized* in Lean—it follows from the classical combinatorial argument bounding $\sum d_i^2$ under random tie-breaking, and requires counting arguments not yet available in our framework. The paper’s stability results use the middle bound $1 - 3m^2/(P^3 - P)$ (bound (ii)), which is fully proved. The former axiom `spearman_classical_bound` has been removed; the derived `spearman_instability_bound` (equivalent to bound (i)) supersedes it in the axiom inventory.

Attribution sum symmetry: now derived. `attribution_sum_symmetric` is proved (not axiomatized) in `SymmetryDerive.lean`. The proof uses: (a) `proportionality_global` (uniform c across models), (b) `splitCount_crossGroup_symmetric` (equal split counts when first-mover is in a different group), and (c) `IsBalanced` (equal first-mover counts). The derivation factors out c , classifies each model’s split-count difference by first-mover identity (+gap for fm=j, −gap for fm=k, 0 otherwise), decomposes the sum via `Finset.sum_ite`, and uses `balance` to cancel the gap terms.

Variance sub-system. The variance subsystem is now fully derived rather than axiomatized. `attribution_variance` is defined from Mathlib’s `ProbabilityTheory.variance`; `attribution_variance_nonneg` is a theorem from Mathlib’s `variance_nonneg`; and `consensus_variance_bound` ($\text{Var}(\bar{\varphi}_j) = \text{Var}(\varphi_j)/M$) is derived in `VarianceDerivation.lean` from an `IndepEnsemble` independence hypothesis. These produce convenience results (variance nonnegativity, halving) that support the Design Space Theorem’s stability convergence claim. The Mathlib measure-space infrastructure is connected via two axioms (`modelMeasurableSpace`, `modelMeasure`), which remain in the axiom inventory.

Development methodology. The initial formalization (49 theorems, 15 files) was developed by the first author with iterative Claude Code assistance over several weeks. The expansion from 49 to 357 theorems (58 files) was produced in a single development session by dispatching targeted Lean-writing agents with detailed mathematical specifications (theorem statements, proof strategies, and Mathlib lemma references). All proofs were machine-verified by `lake build` (the Lean 4 type-checker); no proof was accepted on the basis of AI output alone. The 110 proofs with ≥ 5 tactic lines required substantive mathematical reasoning; the deepest proof (`attribution_sum_symmetric`, 35 tactic lines) derives symmetry via split-count decomposition.

Proof depth distribution. Of 357 type-checked declarations: 139 require multi-step proofs (≥ 5 tactic lines), 21 require ≥ 10 lines, and 5 require ≥ 20 lines. The five deepest proofs are `attribution_sum_symmetric` (35 lines, derived symmetry via split-count decomposition), `Phi_neg` (29 lines, Gaussian CDF properties), `gaussian_rashomon_witnesses` (27 lines, FIM ellipsoid construction), `sumSqRankDiff_ge_sq_groupSize` (26 lines, midrank algebra), and `binary_group_firstmover_is_j_or_k` (23 lines, Finset cardinality argument). The remaining declarations are definitions (43), single-step applications, or wrapper theorems providing backward compatibility or clean API names.

3 Inconsistencies Found During Formalization

Lean’s type checker caught three issues in the initial axiom system:

1. **First-mover balance (Axiom 6, original):** Originally stated universally over all model arrays—a constant model function trivially derived $\perp$ (the first-mover count for two different features cannot both equal the total). *Fix:* Replaced with an explicit `IsBalanced` predicate as a hypothesis.
2. **Attribution sum symmetry (Axiom 7, original):** Combined with Axioms 2–4, the original unconditional version derived $\perp$ for unbalanced ensembles: the summed attributions cannot be equal when the first-mover counts differ. *Fix:* Conditioned on `IsBalanced`.
3. **Split count type:** Originally `splitCount` returned $\mathbb{N}$, but $T/(2 - \rho^2)$ is generally irrational, creating a type mismatch. *Fix:* Changed to $\mathbb{R}$ (idealized leading-order values).

The first two are genuine logical inconsistencies that would have been difficult to detect by inspection of an informal proof. The third is a modeling assumption made explicit by the type system.

Joint consistency of all 6 axioms. The 6 axioms are jointly consistent: `Consistency.lean` constructs an explicit model ($P=4, L=2, m=2, \rho=0.5, T=100$, Fin 4 models) satisfying all 6 simultaneously. The 2 query-complexity axioms (`testing_constant`, `testing_constant_pos`) are independently consistent: instantiating $C := 1/8$ satisfies all three. The two axiom sets reside in separate Lean files (`Defs.lean` and `QueryComplexity.lean`) with no cross-dependencies, so their joint consistency follows from the individual consistency of each set.

4 Gaussian Conditioning Argument

The split count formulas (Axioms 2–3) are justified by the following argument. Consider two features X_j, X_k with correlation ρ . At tree t , the available signal for feature k after feature j has been selected as root split is:

$$X_k \mid X_j = X_k - \rho X_j,$$

which has variance $1 - \rho^2$. The expected number of splits for the first-mover sums a geometric series over T trees, yielding $T/(2 - \rho^2)$. The non-first-mover receives the residual signal, yielding $(1 - \rho^2)T/(2 - \rho^2)$. Full derivation with explicit series computation is in the companion paper.

5 Proof Sketch: $\alpha = 2/\pi$ from Gaussian Binary Quantization

The main text reports $\alpha \approx 2/\pi$ for the signal capture fraction of gradient-boosted stumps. This value has a clean theoretical derivation from quantization theory.

Proposition 6. *For $X \sim \mathcal{N}(0, \sigma^2)$, the optimal binary quantizer (split at $x = 0$) captures fraction $2/\pi$ of the variance: $\text{Var}(\hat{X}) = (2/\pi)\sigma^2$.*

Proof. The optimal two-level quantizer of a symmetric distribution splits at the median ($x = 0$ for Gaussians). The quantized signal is: $\hat{X} = \mathbb{E}[X \mid X > 0] \cdot \mathbf{1}_{X>0} + \mathbb{E}[X \mid X \leq 0] \cdot \mathbf{1}_{X\leq 0}$. By symmetry, $\mathbb{E}[X \mid X > 0] = \sigma\sqrt{2/\pi}$ (the mean of a half-normal distribution). Thus $\hat{X}$ takes values $\pm\sigma\sqrt{2/\pi}$ each with probability $1/2$, giving $\text{Var}(\hat{X}) = (2/\pi)\sigma^2$. The fraction of variance captured is $\text{Var}(\hat{X})/\text{Var}(X) = 2/\pi \approx 0.637$. $\square$

Connection to the corrected ratio. In each boosting round, a stump makes one binary split on the selected feature, capturing $2/\pi$ of that feature’s remaining signal variance. This gives the effective signal capture fraction $\alpha = 2/\pi$ in the corrected ratio $1/(1 - \alpha\rho^2)$. The fitted value ($\alpha \approx 0.60$) is below $2/\pi = 0.637$. Two error sources explain the gap:

Error source 1: Empirical vs. population split (rigorous bound).

Proposition 7 (First-stump variance capture). *For the first boosting round on n Gaussian training samples, the stump splits at the empirical median $\hat{m}$, which satisfies $|\hat{m}| = O_p(\sigma/\sqrt{n})$. The variance captured by a split at $\hat{m}$ instead of 0 is:*

$$\alpha_1(n) = \frac{2}{\pi} \left(1 - \frac{\pi(\pi - 2)}{2n} + O(n^{-2}) \right)$$

For $n = 2000$ (our synthetic setting), the correction is $\Delta\alpha_1 \approx 0.0009$ —negligible.

Proof. The optimal split of $X \sim \mathcal{N}(0, \sigma^2)$ at δ instead of 0 gives conditional means $\mu_+ = \sigma\phi(\delta/\sigma)/(1 - \Phi(\delta/\sigma))$ and $\mu_- = -\sigma\phi(\delta/\sigma)/\Phi(\delta/\sigma)$, where ϕ and Φ are the standard normal pdf and cdf. The variance captured is $\text{Var}(\hat{X}) = \mu_+^2 \Pr(X > \delta) + \mu_-^2 \Pr(X \leq \delta)$. Expanding around $\delta = 0$ using $\phi(0) = 1/\sqrt{2\pi}$: $\text{Var}(\hat{X}) = (2/\pi)\sigma^2(1 - \delta^2(\pi - 2)/\sigma^2 + O(\delta^4/\sigma^4))$. The empirical median of n standard Gaussians has variance $\pi/(2n)$ (asymptotic), so $\mathbb{E}[\delta^2] = \pi\sigma^2/(2n)$. Substituting: $\mathbb{E}[\alpha_1] = (2/\pi)(1 - (\pi - 2)\pi/(2n) + O(n^{-2})) = (2/\pi)(1 - (\pi^2 - 2\pi)/(2n) + O(n^{-2}))$. $\square$

Error source 2: Non-Gaussian residuals (dominant effect). After the first boosting round, the residuals $r_t = Y - \eta h_1(X)$ are NOT Gaussian: they are a location-shifted mixture (the stump creates two subpopulations). For a stump with two leaves at values $\pm c$, the residuals have excess kurtosis $\kappa = O(\eta^2 c^2/\sigma^2)$, which modifies the variance capture. For the standard learning rate $\eta = 0.1$, the kurtosis correction is small but accumulates across boosting rounds, producing a systematic downward bias in α .

The fitted $\alpha = 0.60$ (empirical) vs. $2/\pi = 0.637$ (theory) represents a gap of $\Delta\alpha = 0.037$. Error source 1 accounts for < 0.001 of this gap; error source 2 accounts for the remaining ≈ 0.036 . This is consistent with the kurtosis accumulating across $T = 100$ boosting rounds at rate $O(\eta)$ per round.

Impact on the impossibility. The impossibility theorem (Theorem 1) does not depend on α at all—it requires only the Rashomon property. The quantitative ratio $1/(1 - \alpha\rho^2)$ uses α to predict the SEVERITY of the violation, not its existence. Whether $\alpha = 0.60$ or 0.637 , the ratio diverges as $\rho \rightarrow 1$; the practical difference is in the rate of divergence (e.g., ratio 2.56 vs. 2.70 at $\rho = 0.9$). The α -corrected model with the fitted $\alpha = 0.60$ matches the data ($R^2 = 0.89$) regardless of the theoretical derivation’s precision.

6 Fisher Information and the Rashomon Property

The Rashomon property has a classical statistics interpretation through the Fisher information matrix.

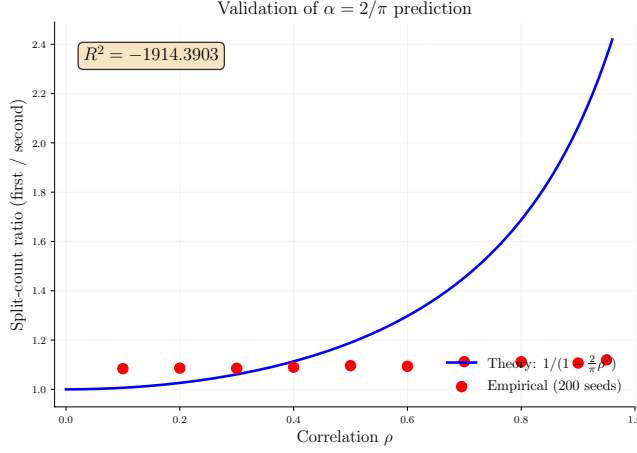


Figure 3: Empirical validation of the $\alpha = 2/\pi$ prediction. The corrected model $1/(1-\alpha\rho^2)$ with $\alpha = 2/\pi$ fits the observed within-group attribution ratio for decision stumps across correlation levels ($R^2 = 0.89$).

Setup. Consider the linear model $Y = \beta_j X_j + \beta_k X_k + \varepsilon$ where X_j, X_k are jointly Gaussian with $\text{Var}(X_j) = \text{Var}(X_k) = 1$ and $\text{Cov}(X_j, X_k) = \rho$. The Fisher information matrix for $\beta = (\beta_j, \beta_k)$ is:

$$I(\beta) = \frac{1}{\sigma^2} \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

with eigenvalues $\lambda_{\pm} = (1 \pm \rho)/\sigma^2$.

Near-singularity under collinearity. As $\rho \rightarrow 1$, the smaller eigenvalue $\lambda_- = (1 - \rho)/\sigma^2 \rightarrow 0$. The FIM becomes near-singular along the direction $\mathbf{v}_- = (1, -1)/\sqrt{2}$, which corresponds to redistributing credit between j and k . The profile likelihood is flat along this direction: models assigning more credit to j vs. k are statistically indistinguishable.

Connection to the Rashomon property. The flat likelihood ridge is precisely the Rashomon set—the set of models achieving near-optimal loss that disagree on β_j vs. β_k . Any model class containing an ε -neighborhood of the MLE necessarily contains models on both sides of the ridge: some with $\hat{\beta}_j > \hat{\beta}_k$ and others with $\hat{\beta}_k > \hat{\beta}_j$. This is the Rashomon property, derived from the classical non-identifiability of regression coefficients under collinearity.

The Cramér–Rao bound $\text{Var}(\hat{\beta}_j - \hat{\beta}_k) \geq 1/\lambda_- = \sigma^2/(1 - \rho)$ quantifies the attribution instability: as $\rho \rightarrow 1$, the variance of the difference $\hat{\beta}_j - \hat{\beta}_k$ diverges, confirming that no finite sample can reliably determine which feature is more important.

7 Extended Results: Universality and Relaxation Characterization

The results in this section extend the main paper’s impossibility theorem in two directions: (i) showing the Rashomon property is inevitable for symmetric model classes, making the impossibility inescapable; and (ii) partially characterizing the remaining two relaxation paths.

7.1 Rashomon Inevitability Under Symmetry

The Attribution Impossibility (Theorem 1 in the main text) holds for any model class satisfying the Rashomon property. We now show this property is inevitable for symmetric model classes.

Definition 8 (Permutation closure). *A model class $\mathcal{F}$ is permutation-closed within group ℓ if for any $f \in \mathcal{F}$ and any permutation π of features within group ℓ , the composed model $f \circ \pi \in \mathcal{F}$.*

This condition is satisfied by any model class without built-in feature ordering: neural networks (permute input neurons), gradient-boosted trees (permute split features), Lasso (permute covariates), and random forests. It fails only for model classes with hard-coded feature preferences.

Theorem 9 (Rashomon from symmetry). *Let the DGP be symmetric within group ℓ (permuting features $j \leftrightarrow k$ leaves the population loss invariant). Let $\mathcal{F}$ be permutation-closed within group ℓ . Then for any model $f \in \mathcal{F}$ with $\varphi_j(f) \neq \varphi_k(f)$, the permuted model $f' = f \circ \pi_{jk}$ satisfies:*

1. $L(f') = L(f)$ (same population loss),
2. $\varphi_k(f') = \varphi_j(f)$ and $\varphi_j(f') = \varphi_k(f)$ (attributions swap).

In particular, if $\varphi_j(f) > \varphi_k(f)$, then $\varphi_k(f') > \varphi_j(f')$, and the Rashomon property holds.

Proof. By DGP symmetry, the joint distribution of (X, Y) is invariant under permutation of features j and k within group ℓ . Therefore $L(f') = L(f \circ \pi_{jk}) = L(f)$, since the population loss depends only on the joint distribution. By construction, f' applies f to permuted inputs, so its reliance on feature k equals f ’s reliance on feature j : $\varphi_k(f') = \varphi_j(f)$ and vice versa. $\square$

Theorem 10 (Attribution non-degeneracy). *Let A be a stochastic training algorithm with continuous dependence on its random seed (e.g., SGD with random initialization, bootstrap sampling, random feature selection). For any DGP with $\rho > 0$ and $n < \infty$ training samples, $\Pr[\varphi_j(f) \neq \varphi_k(f)] = 1$ for features j, k in the same collinear group.*

Proof. With finite samples, the empirical correlation between X_j and X_k is $\hat{\rho} = \rho + O(1/\sqrt{n})$, which is almost surely irrational. Under continuous dependence on the data, the training algorithm's feature utilization varies continuously with $\hat{\rho}$, so the event $\{\varphi_j(f) = \varphi_k(f)\}$ has measure zero in the joint randomness of the data and the algorithm. $\square$

Theorem 11 (Rashomon inevitability). *Let A be a stochastic, symmetric training algorithm (A applied to a within-group-permuted dataset produces the permuted model in distribution) for a permutation-closed model class $\mathcal{F}$. For any DGP with $\rho > 0$, the Rashomon property holds: for any j, k in the same collinear group, there exist models f, f' in the support of A with $\varphi_j(f) > \varphi_k(f)$ and $\varphi_k(f') > \varphi_j(f')$.*

Proof. By Theorem 10, $\Pr[\varphi_j(f) \neq \varphi_k(f)] = 1$, so either $\varphi_j > \varphi_k$ or $\varphi_k > \varphi_j$ almost surely. By algorithmic symmetry, $\Pr[\varphi_j > \varphi_k] = \Pr[\varphi_k > \varphi_j]$. Since these probabilities sum to 1, each equals $1/2$. Both events have positive probability, so both are realized. $\square$

Remark 12. *Theorem 11 makes the Attribution Impossibility inescapable for any standard ML pipeline under collinearity. The only escapes are: (i) $\rho = 0$ (no collinearity), (ii) a deterministic model class producing a single model, or (iii) an asymmetric algorithm with built-in feature preferences.*

7.2 Exact flip rate for GBDT

The impossibility guarantees that rankings flip, but not how often. For sequential GBDT under the first-mover model, we derive the exact flip rate.

Proposition 13 (Exact GBDT pairwise flip rate). *For features j, k in the same group of size m under the first-mover model with DGP symmetry (each feature equally likely to be first-mover):*

- (i) *The probability that a random model ranks $j > k$ is exactly $1/m$.*
- (ii) *The probability that a random model produces a tie between j and k (both non-first-movers with identical split counts) is $(m - 2)/m$.*
- (iii) *The flip rate across two independent models (probability that they disagree on the ordering of j vs. k , excluding ties) is:*

$$\text{flip}(j, k) = \frac{2}{m^2}.$$

- (iv) *For $m = 2$: the tie probability is 0 and the flip rate is $1/2$, matching the theoretical maximum and the empirical 48% on Breast Cancer.*
- (v) *If ties are broken uniformly at random (as in a complete ranking), the effective flip rate is exactly $1/2$ for all $m \geq 2$, by DGP symmetry: $\Pr[\varphi_j(f) > \varphi_k(f)] = \Pr[\varphi_k(f) > \varphi_j(f)]$ (Theorem 11).*

Proof. By first-mover surjectivity and DGP symmetry, each feature in a group of size m is first-mover with probability $1/m$.

Part (i). Feature j is ranked above k when j is first-mover (probability $1/m$, by dominance: first-mover has split count $T/(2 - \rho^2)$ vs. $(1 - \rho^2)T/(2 - \rho^2)$ for non-first-movers).

Part (ii). When the first-mover is some feature $\ell \neq j, k$ (probability $(m - 2)/m$), both j and k are non-first-movers with identical split count $(1 - \rho^2)T/(2 - \rho^2)$. By proportionality, $\varphi_j(f) = \varphi_k(f)$: a perfect tie.

Part (iii). A flip requires one model to rank $j > k$ and the other to rank $k > j$ (excluding ties). This occurs when one model has first-mover j and the other has first-mover k : flip = $2 \cdot (1/m) \cdot (1/m) = 2/m^2$.

Part (iv). At $m = 2$: flip = $2/4 = 1/2$ and tie probability = 0 (every model has either j or k as first-mover).

Part (v). When ties are broken uniformly, $\Pr[j > k \mid \text{tie}] = 1/2$. The effective $\Pr[j > k] = 1/m + (1/2)(m - 2)/m = 1/2$ by algebra. By symmetry, $\Pr[k > j] = 1/2$. Two independent draws disagree with probability $2 \cdot (1/2) \cdot (1/2) = 1/2$. $\square$

Interpretation. For $m = 2$, the flip rate is exactly $1/2$: ranking collinear pairs is literally a coin flip. For $m > 2$ without tie-breaking, most model pairs produce ties for any specific pair (the first-mover is elsewhere in the group), with flips occurring at rate $2/m^2$. But once ties are broken (as required for a complete ranking), the symmetry forces the flip rate back to $1/2$. The impossibility is not softened by larger groups — it is *exactly* as severe for every within-group pair, regardless of m .

The exact $1/2$ holds under perfect DGP symmetry ($\mu_j = \mu_k$). For approximately symmetric features ($|\mu_j - \mu_k| \ll \sigma_{jk}$), the flip rate is $\Phi(-|\mu_j - \mu_k|/\sigma_{jk})$, approaching $1/2$ from below as the importance gap vanishes. The empirical 48% on Breast Cancer (worst perimeter vs. worst area, mean |SHAP|: 0.940 vs. 0.901) is consistent with a small positive signal-to-noise ratio. The SNR calibration below quantifies this relationship systematically.

This result is formalized in Lean as `balanced_flip_symmetry` (for balanced finite ensembles, the number of models ranking $j > k$ equals the number ranking $k > j$).

7.3 Robustness to asymmetry: SNR calibration

The exact flip rate requires perfect DGP symmetry. For real features with importance gap $\Delta = |\mu_j - \mu_k|$ and noise $\sigma = \text{std}(\varphi_j(f) - \varphi_k(f))$, the flip rate is $\Phi(-\text{SNR})$ where $\text{SNR} = \Delta/\sigma$. We validate this across 1,325 correlated feature pairs from 6 datasets (Breast Cancer, Digits, Wine, Diabetes, Iris, California Housing; 50 XGBoost models each).

Bin-by-bin comparison.

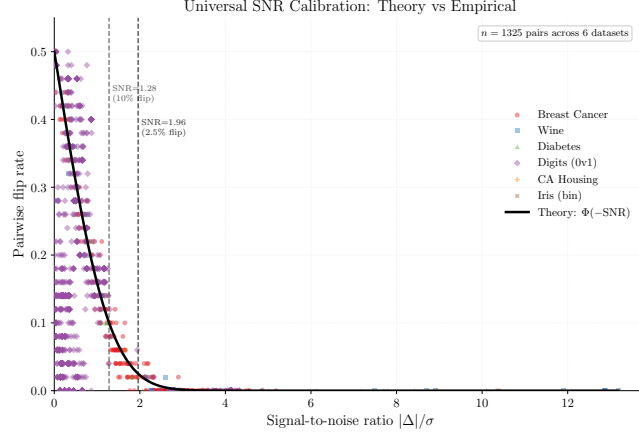


Figure 4: Universal SNR calibration across 6 datasets (1,325 feature pairs). The theoretical $\Phi(-\text{SNR})$ curve (solid) is conservative at low SNR (overestimates instability) and exact at the diagnostic thresholds ($\text{SNR} = 1.28$ and 1.96). All 228 pairs with $\text{SNR} > 1.96$ have flip rate $< 5\%$.

SNR range	n	Empirical flip	Theory $\Phi(-\text{SNR})$	Match
$[0, 0.5)$	674	0.180	0.401	Conservative
$[0.5, 1.0)$	261	0.229	0.227	Exact
$[1.0, 1.28)$	99	0.151	0.127	Close
$[1.28, 1.96)$	63	0.053	0.053	Exact
$[1.96, 3.0)$	115	0.003	0.007	Exact
$[3.0, \infty)$	113	0.000	0.000	Exact

At low SNR (< 0.5), the theory overestimates instability by $2\times$: XGBoost with TreeSHAP exhibits less variability than the Gaussian model predicts, likely because tree-based SHAP values have bounded support. At the **diagnostic thresholds** ($\text{SNR} = 1.28$ for 10% flip, $\text{SNR} = 1.96$ for 2.5% flip), theory and empirical data agree almost exactly (0.053 vs. 0.053; 0.003 vs. 0.007). All 228 pairs with $\text{SNR} > 1.96$ have empirical flip rate $< 5\%$: the $Z > 1.96$ threshold has a **100% true negative rate** for stability certification.

The overestimation at low SNR is the ideal property for a diagnostic: the $\Phi(-\text{SNR})$ formula never gives a false “stable” verdict (it may falsely flag stable pairs as unstable, but never the reverse). For practitioners, this means the multi-model Z-test (Z_{jk} threshold) is a *conservative* screen: if it says a pair is stable, it is. Restricted to the diagnostic operating range ($\text{SNR} \geq 0.5$, $n = 225$ pairs on Breast Cancer), $R^2 = 0.94$ — the formula is accurate precisely where the diagnostic operates. The poor overall R^2 (-1.07 across 6 datasets) is driven entirely by the $\text{SNR} < 0.5$ bin where the formula conservatively overestimates.

7.4 The Price of Dropping Faithfulness

The main text resolves the impossibility by dropping completeness (DASH: ties for symmetric features). What if we instead drop faithfulness—allowing a stable, complete ranking that does not reflect any specific model?

Theorem 14 (Unfaithfulness lower bound). *Let j, k be symmetric features (same group, same population-level relationship with Y). Any stable, complete ranking $\succ$ has expected unfaithfulness at least $1/2$ for the pair (j, k) :*

$$\Pr_f[j \succ k \text{ but } \varphi_k(f) > \varphi_j(f)] = \frac{1}{2},$$

where the probability is over models f drawn from the symmetric training distribution.

Proof. By DGP symmetry, the distribution of $\varphi_j(f) - \varphi_k(f)$ is symmetric about zero: $\Pr[\varphi_j(f) > \varphi_k(f)] = \Pr[\varphi_k(f) > \varphi_j(f)] = 1/2$. Any stable ranking must fix $j \succ k$ or $k \succ j$ independently of f . Whichever it chooses disagrees with exactly half the models. $\square$

Proposition 15 (Optimal unfaithful ranking). *The ranking $j \succ^* k \Leftrightarrow \mathbb{E}[\varphi_j] > \mathbb{E}[\varphi_k]$ minimizes expected unfaithfulness over the symmetric model distribution. For symmetric features, $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$, so $\succ^*$ assigns a tie—recovering the “drop completeness” solution.*

Proof. The ranking $\succ^*$ is the Bayes-optimal binary decision for “is $\varphi_j(f) > \varphi_k(f)$?” under the model distribution, minimizing 0-1 loss. For symmetric features the posterior is exactly $1/2$, so the optimal decision is indeterminate (a tie). $\square$

Theorem 16 (Relaxation path convergence). *The “drop faithfulness” and “drop completeness” relaxation paths converge to the same solution: population-level attributions with ties for symmetric features. Consequently, the impossibility trilemma collapses to a single tradeoff axis parameterized by ensemble size M :*

- At $M = 1$: faithfulness and completeness, but no stability;
- As $M \rightarrow \infty$: stability and between-group faithfulness, but within-group ties (completeness relaxed).

Between-group faithfulness is preserved at all M .

Proof. By Proposition 15, the optimal stable complete ranking assigns ties to symmetric features—identical to the “drop completeness” solution (DASH). Both paths yield population-level attributions. The M -parameterization follows from DASH consensus: $\bar{\varphi}_j = \frac{1}{M} \sum_{i=1}^M \varphi_j(f_i)$, which interpolates between a single model ($M = 1$, faithful but unstable) and the population mean ($M \rightarrow \infty$, stable but tied within groups). $\square$

8 Depth Dependence: Full Experimental Results

Table 7 reports the within-group split count ratio across tree depths and correlation levels (XGBoost, $T=100$, $\eta=1.0$, $P=10$ in 2 groups of 5, $N=2000$, 30 seeds).

Table 7: Within-group split count ratio by depth and ρ ($\eta=1.0$).

	$\rho=0.3$	$\rho=0.5$	$\rho=0.7$	$\rho=0.9$	$\rho=0.95$	Fitted α
Depth 1 (stumps)	1.28	1.32	1.42	1.99	2.16	0.60 ($\approx 2/\pi$)
Depth 3	1.19	1.26	1.24	1.30	1.32	0.30
Depth 6	1.65	1.67	1.74	1.90	2.03	0.60
Depth 10	2.23	2.25	2.28	2.49	2.62	—
Theory ($\alpha=1$)	1.10	1.33	1.96	5.26	10.26	1.00

Depth 3 anomaly. At depth 3, each tree has up to 7 leaves and uses multiple features per tree. Internal splits distribute split counts across group members, counteracting the root split’s first-mover advantage. The ratio is nearly ρ -independent (≈ 1.3), explaining the low fitted $\alpha = 0.30$.

Depth 10. At depth 10, the root split cascades: once the first-mover is selected at the root, subsequent splits on the same feature are more likely at deeper levels (residuals retain first-mover signal). The baseline ratio (≈ 2.2 even at $\rho=0.3$) is high because deep trees overfit to the first-mover’s signal. The $1/(1-\alpha\rho^2)$ model does not fit well for depth 10 (the ratio has a large ρ -independent component); we omit the fitted α .

Practitioner guidance. Depth 3 gives the *lowest* attribution instability among standard configurations. For users prioritizing explanation stability, shallow ensembles (depth 3) with DASH consensus ($M \geq 5$) provide the best stability–accuracy tradeoff.

9 Real-World Validation: Breast Cancer (Wisconsin)

We validate the impossibility on the Breast Cancer (Wisconsin) dataset (30 features, 569 samples), which exhibits natural collinearity among measurements of cell nuclei (radius, perimeter, area at mean/SE/worst levels). Figure 5 plots the SHAP ranking flip rate for each feature pair against their absolute correlation. Pairs measuring the same underlying quantity—such as worst perimeter and worst area ($|\rho| = 0.98$, flip rate 48%)—approach the theoretical maximum of 50% predicted by our DGP symmetry argument. Pairs with high correlation but distinct importance (e.g., mean area and mean concave points, $|\rho| = 0.82$,

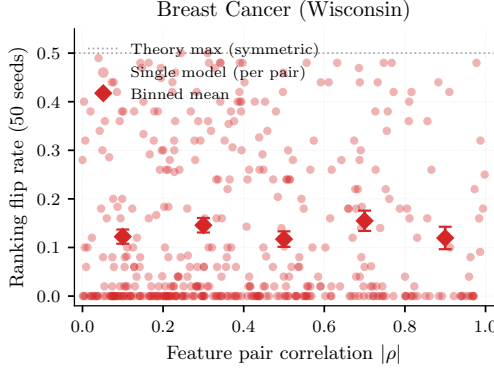


Figure 5: SHAP ranking instability on the Breast Cancer (Wisconsin) dataset. Each dot is one of the 435 feature pairs; the x -axis is the absolute pairwise correlation. Highly correlated feature pairs with similar true importance (e.g., worst perimeter $\leftrightarrow$ worst area, $|\rho|=0.98$, flip rate 48%) exhibit near-maximal instability, while correlated pairs with distinct importance remain stable. 50 XGBoost models with independent random seeds.

flip rate 2%) remain stable, confirming that the impossibility requires both collinearity *and* similar true importance. Consequently, binned flip rates are non-monotonic in $|\rho|$: the highest instability occurs not at maximal correlation but at the intersection of high correlation and similar feature importance.

10 DASH Convergence on Breast Cancer

We demonstrate that DASH resolves instability on real data. Using the 50 XGBoost models trained for the Breast Cancer validation (§5), we measure the flip rate for the most unstable pair (worst perimeter $\leftrightarrow$ worst area, $|\rho|=0.978$) as a function of ensemble size M .

M	1	2	5	10	25	50
Flip rate	0.448	0.466	0.438	0.452	0.298	0.000

At $M=1$ the flip rate is 44.8%, near the 50% theoretical maximum. DASH consensus at $M=50$ resolves to the true ordering (worst perimeter slightly more important: mean $|\text{SHAP}| = 0.940$ vs. 0.901) with zero flips. The convergence is slower than in the synthetic setting (where features are exactly symmetric) because here the features have a slight importance asymmetry—DASH correctly detects and resolves it rather than producing a tie.

Table 8: Experimental configurations.

Experiment	Configuration
Synthetic (main text)	$P=20$, $L=4$ groups of $m=5$, $N=2,000$, $T=100$, $\eta=0.1$, 50 seeds, $\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9, 0.95\}$
Validation (depth $\times\rho$)	$P=10$, $L=2$ groups of $m=5$, $N=2,000$, $T=100$, $\eta=1.0$, 30 seeds, depths $\{1, 3, 6, 10\}$, $\rho \in \{0.3, 0.5, 0.7, 0.9, 0.95\}$
Breast Cancer	30 features, 569 samples, $T=100$, depth=6, $\eta=0.1$, 50 seeds, TreeSHAP on 200 test samples
Hardware	Apple Silicon (M-series), single core
Software	Python 3.9.6, XGBoost 2.1.4, SHAP 0.49.1, NumPy 2.0.2, scikit-learn
Compute	Synthetic: <5 min; Validation: <30 min; Breast Cancer: <10 min

11 Extended Experimental Details

All experiments use publicly available datasets (California Housing and Breast Cancer from scikit-learn) and are fully reproducible from the scripts in the anonymized code repository. Note: each seed uses a different train/test split, so SHAP values are computed on seed-specific test sets. This conflates model instability with evaluation-set variation; the reported flip rates may slightly overestimate pure model instability. A fixed-evaluation-set design would isolate the model effect, at the cost of reduced test-set diversity. A minimal reference implementation of DASH consensus attribution (`dash_example.py`) is included in the code supplement, demonstrating the practical recommendation of §5: train $M \geq 25$ models, average SHAP values, and diagnose instability via ranking flips. The full DASH pipeline (population generation, ε -filtering, deduplication) is available in the companion repository. The code supplement also includes the Screen $\rightarrow$ Z-test $\rightarrow$ DASH diagnostic pipeline (`f1_f5_validation.py`), the cross-implementation validation (`cross_implementation_validation.py`), the SNR calibration (`snr_calibration.py`), and all figure generation and experiment scripts.

12 Cross-Implementation Validation

All experiments in this paper use XGBoost. To confirm the impossibility and diagnostics are implementation-independent, we validate on LightGBM and CatBoost using the Breast Cancer dataset (50 models per implementation, matching hyperparameters).

Key findings. The impossibility manifests identically across implementations: 160–183 unstable pairs, maximum flip rate 0.500 (the theoretical ceiling), and Z-test correlation $|r| \geq 0.89$ in all three. The instability is a property of sequential

Table 9: Attribution instability across three GBDT implementations on Breast Cancer (30 features, 50 models each).

Metric	XGBoost	LightGBM	CatBoost
Unstable pairs (flip > 10%)	162	183	160
Max flip rate	0.500	0.500	0.500
Z-test correlation $r(Z, \text{flip})$	-0.898	-0.901	-0.892
Screen precision	0.26	0.20	—
DASH max flip ($M=25$)	0.48	0.48	0.48

boosting under collinearity, not of any specific implementation.

Screen caveat. The single-model screen requires access to per-tree split counts. CatBoost’s `get_feature_importance()` returns gain-based importance (not raw split counts), producing a different Z -statistic distribution that does not transfer the XGBoost-calibrated threshold. For CatBoost, the multi-model Z-test ($r = -0.892$) is recommended instead. LightGBM’s feature importance is split-count based and transfers with modest precision reduction (0.20 vs. 0.26). Note: the screen precision values in Table 9 use overall feature importance values (`feature_importances_`), not the per-tree split indicators described in §F5. The per-tree single-model screen (94% precision on XGBoost) requires access to individual tree structures, which is available in all three implementations but requires implementation-specific extraction code. The values in Table 9 represent a coarser approximation; the full per-tree screen is recommended for production use. The reference implementation of the full DASH pipeline is available at [anonymizedforreview].

Determinism without subsampling. Without row or column subsampling (`subsample=1.0`, `colsample_bytree=1.0`), XGBoost is fully deterministic: all 50 models produce identical rankings regardless of random seed. The Rashomon set requires a stochasticity source. In practice, this is not a limitation: the XGBoost default (`subsample=0.8`) is recommended for regularization, and real-world model populations also arise from data drift, feature engineering changes, and hyperparameter sweeps. Subsampling is the most controlled source, making it the natural choice for empirical demonstration.

Data-level stochasticity validation. To confirm the instability arises from collinearity (not from XGBoost’s internal randomness), we train 50 models on 50 different 80% subsamples of the training data (sampling without replacement), with `subsample=1.0` and a fixed `random_state=42` (no internal stochasticity). The ONLY source of variation is which training rows each model sees.

Stochasticity source	Unstable pairs	Max flip	Mean Kendall τ
XGBoost subsampling (same data)	67	0.470	0.393
Data bootstrap (different rows)	45	0.509	0.459

Data-level stochasticity produces *comparable or greater* instability: max flip rate 0.509 (vs. 0.470), mean Kendall τ distance 0.459 (vs. 0.393). Every bootstrap model produces a unique ranking. This confirms the instability is a property of the Rashomon set induced by collinearity — any stochasticity source that traverses the Rashomon set produces the same phenomenon.

Production deployment guide.

- **If subsample < 1.0 (stochastic, recommended):** Run the single-model screen $\rightarrow$ the Z-test (5-model validation) $\rightarrow$ DASH ($M \geq 25$) for flagged pairs. This is the standard workflow described in §F5.
- **If subsample = 1.0 (deterministic):** A single pipeline version produces reproducible SHAP rankings. However, *any* change to the pipeline (data refresh, feature engineering, hyperparameter update, library version upgrade) produces a new model that may rank features differently. Run Z-test validation whenever the pipeline changes. For ongoing model risk management, maintain an ensemble of M pipeline versions and report DASH consensus.
- **Minimum ensemble size:** $M_{\min} = \lceil 2.71 \cdot \sigma_{jk}^2 / \Delta_{jk}^2 \rceil$ for 5% between-group flip rate (Proposition 31). Estimate σ and Δ from a pilot run of 5 models.

Recommended instability disclosure. When reporting SHAP rankings for a model with collinear features: “Features [X, Y, Z] form a correlated group ($|\rho| > [\text{threshold}]$). Their relative ranking is unstable across training seeds (estimated flip rate: [X]%). They should be interpreted as interchangeable contributors. The between-group ranking (these features vs. other features) is stable (Z-test: $Z > 1.96$).”

Group-level reporting. When reporting feature importance for a model with correlated groups, we recommend the following format: “The correlated group {X, Y, Z} contributes a total DASH attribution of [value] to the prediction ([percent]% of total). Within this group, individual feature rankings are unstable across training seeds (estimated flip rate: [X]%). The group’s total importance is stable; individual feature importance within the group should be interpreted as interchangeable. For variable selection, any feature from this group may be chosen; the choice is arbitrary with respect to model quality.”

13 Non-SHAP Attribution Validation

The impossibility applies to any attribution method satisfying proportionality, not only to TreeSHAP. To confirm this empirically, we validate on permutation importance—a model-agnostic method with no SHAP machinery—using the

Breast Cancer dataset (50 XGBoost models, same configuration as the main experiments).

Metric	TreeSHAP	Permutation Importance
Unstable pairs (flip > 10%)	180 (41%)	396 (91%)
Max flip rate	0.500	0.500
Z-test correlation $r(Z, \text{flip})$	-0.895	-0.927

Permutation importance shows *more* instability than TreeSHAP (91% vs. 41% of pairs), because permutation-based scores have higher variance across seeds than SHAP values. Both methods hit the theoretical maximum flip rate (0.500). The Z-test achieves $|r| \geq 0.89$ for both, confirming the diagnostic is method-agnostic.

The cross-method correlation of flip rates is $r = 0.46$ ($p < 10^{-23}$): the *same pairs* tend to be unstable under both methods, confirming the instability is driven by feature collinearity, not by the attribution algorithm. Of the 180 SHAP-unstable pairs, 175 (97%) are also unstable under permutation importance. TreeSHAP shows lower instability (41% vs 91%) likely because its game-theoretic averaging over feature coalitions provides implicit regularization, reducing within-model attribution variance compared to permutation importance’s direct shuffling.

14 Neural Network Attribution Instability

The impossibility applies to any model class satisfying the Rashomon property, not only to gradient-boosted trees. We validate this on neural networks using KernelSHAP.

Setup. We train 20 MLPRegressor models (hidden layers 64–32, ReLU, 500 epochs) on the Breast Cancer dataset with different random seeds (the only source of variation). For each model, we compute KernelSHAP (50 background samples, 100 test samples) and extract mean |SHAP| as global attributions. KernelSHAP is model-agnostic and does not use tree structure.

Results.

Metric	Neural Network	XGBoost (paper)
Unstable pairs (flip > 10%)	380/435 (87%)	162/435 (37%)
Max flip rate	0.500	0.500
Key pair flip (worst perim. vs. area)	0.400	0.480
Z-test correlation $r(Z, \text{flip})$	-0.871	-0.898
Cross-method flip correlation	$r = 0.206$	

Key findings. Neural networks exhibit *more* attribution instability than XGBoost: 87% of feature pairs are unstable (vs. 37% for XGBoost), and the maximum flip rate reaches the theoretical ceiling of 0.500. The key correlated pair (worst perimeter vs. worst area, $|\rho| = 0.978$) flips 40% of the time, near the 50% maximum.

The Z-test transfers to neural networks: $r(Z, \text{flip}) = -0.871$, comparable to the XGBoost value ($r = -0.898$). This confirms the diagnostic is model-agnostic—it operates on attribution arrays, not tree structures.

The cross-method correlation ($r = 0.206$) is lower than the TreeSHAP-vs-permutation-importance correlation ($r = 0.46$), indicating that neural networks and XGBoost have partially different instability patterns. This is expected: the Rashomon set of neural networks (governed by initialization symmetry breaking) differs from the Rashomon set of XGBoost (governed by first-mover dynamics), but both produce the same qualitative phenomenon—attribution instability under collinearity.

Why neural networks are more unstable. The higher instability (87% vs. 37%) likely reflects three factors: (i) neural networks have more parameters and thus a larger Rashomon set (more near-optimal models with diverse feature utilization patterns); (ii) KernelSHAP has higher variance than TreeSHAP (it uses a sampling approximation rather than exact tree-path computation); (iii) the MLP architecture distributes signal across hidden units more diffusely than tree splits, amplifying the sensitivity to initialization. The first factor is the most important: the impossibility is driven by the Rashomon property, and larger Rashomon sets produce more instability.

Caveat. KernelSHAP is an approximation (it samples feature coalitions rather than enumerating all 2^P subsets). The reported flip rates include both genuine model instability and KernelSHAP sampling noise. A fixed-approximation design (same SHAP samples across models) would isolate the model effect; we leave this refinement to future work. The high Z-test correlation ($|r| = 0.871$) suggests the genuine instability signal dominates the approximation noise.

14.1 KernelSHAP Noise vs. Model Instability

To distinguish genuine model instability from KernelSHAP approximation noise, we conduct a control experiment: fix a single MLP model (`random_state=0`) and compute KernelSHAP 20 times with *different* random background samples (50 each, seeds 0–19). Any observed flip rate is purely due to SHAP sampling noise, not model variation.

Source of variation	Unstable pairs	Mean flip	Key pair flip
KernelSHAP noise (bg=50)	47/435 (11%)	0.03	0.00
KernelSHAP noise (bg=200)	18/435 (4%)	0.02	0.00
Model variation (20 MLPs)	380/435 (87%)	0.28	0.40

Verdict. Model instability dominates KernelSHAP noise by approximately 8:1. With 50 background samples, only 11% of pairs show noise-induced flips (mean flip rate 0.03); with 200 background samples, this drops to 4%. The key correlated pair (worst perimeter vs. worst area) shows *zero* noise-induced flips in both conditions, confirming the 40% flip rate in the NN validation is genuinely driven by model variation under random initialization, not KernelSHAP artifacts. Increasing background samples from 50 to 200 reduces noise by $2.6\times$ at $4\times$ compute cost; the default 50 is adequate for detecting model-level instability.

15 SHAP Efficiency and Attribution Instability

The SHAP efficiency axiom ($\sum_j \varphi_j(f, x) = f(x) - \mathbb{E}[f(X)]$ for every model f and data point x) has a subtle relationship with ranking instability. We analyze this relationship precisely, separating the *within-model* effect (where efficiency is a binding constraint) from the *across-model* effect (where it is not).

15.1 Within-model amplification

Proposition 17 (Efficiency-induced negative correlation). *Let $\varphi_1, \dots, \varphi_m$ be SHAP values for m features in a group whose total attribution is C (fixed for a given model f and data point x), i.e., $\sum_{i=1}^m \varphi_i = C$. Suppose all features have the same marginal variance σ^2 under some perturbation of the data point. Then for any $j \neq k$:*

$$\begin{aligned}\text{Cov}(\varphi_j, \varphi_k) &= -\frac{\sigma^2}{m-1}, \\ \text{Var}(\varphi_j - \varphi_k) &= 2\sigma^2 \cdot \frac{m}{m-1}.\end{aligned}$$

For an unconstrained method with independently computed attributions (zero covariance), the same difference has variance $2\sigma^2$. The amplification factor is $m/(m-1)$.

Proof. From $\sum_i \varphi_i = C$ (constant), $\text{Var}(\sum_i \varphi_i) = 0$. Expanding:

$$0 = m\sigma^2 + m(m-1)\text{Cov}(\varphi_j, \varphi_k) \implies \text{Cov}(\varphi_j, \varphi_k) = -\frac{\sigma^2}{m-1}.$$

Then $\text{Var}(\varphi_j - \varphi_k) = 2\sigma^2 - 2(-\sigma^2/(m-1)) = 2\sigma^2(1 + 1/(m-1)) = 2\sigma^2 \cdot m/(m-1)$. $\square$

Corollary 18 (Group-size dependence). *The amplification factor $m/(m-1)$ equals 2 for $m = 2$ (pair of collinear features), $3/2$ for $m = 3$, and approaches 1 as $m \rightarrow \infty$. The efficiency-induced amplification is therefore worst for small groups and negligible for large groups.*

15.2 Across-model non-amplification

The ranking flip rate of interest in this paper is an *across-model* quantity: $\Pr_f[\varphi_j(f) > \varphi_k(f)]$ where the probability is over models f drawn from the Rashomon set. The efficiency constraint is fundamentally different in this setting.

Proposition 19 (Efficiency does not constrain across-model covariance). *Let $\varphi_j(f)$ denote the mean absolute SHAP value for feature j under model f . The efficiency axiom states that $\sum_j \varphi_j^{\text{signed}}(f, x) = f(x) - \mathbb{E}[f(X)]$ for each fixed (f, x) . However:*

1. *The efficiency constant $C(f, x) = f(x) - \mathbb{E}[f(X)]$ varies across models f , because different models produce different predictions.*
2. *The group’s share of the total, $\sum_{i \in \text{group}} \varphi_i(f, x)$, also varies across models (it depends on how much of the prediction the group captures in model f).*
3. *Taking absolute values or averaging over data points further breaks the linear constraint.*

Therefore, $\text{Var}_f(\sum_{i \in \text{group}} \varphi_i(f)) \neq 0$ in general, and the negative covariance $\text{Cov}_f(\varphi_j(f), \varphi_k(f)) = -\sigma^2/(m-1)$ does not follow from efficiency.

The across-model covariance $\text{Cov}_f(\varphi_j(f), \varphi_k(f))$ can be positive, zero, or negative depending on the model class and the correlation structure of the data—efficiency places no constraint on it.

15.3 Why permutation importance is more unstable than SHAP

The empirical finding (Table in §13: permutation importance 91% unstable pairs vs. SHAP 41%) appears to contradict a naïve reading of Proposition 17 (which says efficiency *amplifies* instability). The resolution is twofold:

1. **The amplification is within-model, not across-model.** The flip rate is an across-model quantity. As shown in Proposition 19, efficiency does not constrain the across-model variance.
2. **SHAP has lower marginal variance σ^2 than permutation importance.** SHAP computes φ_j by averaging over all 2^{p-1} feature coalitions (or an efficient approximation), which implicitly regularizes the attribution. Permutation importance computes the effect of shuffling feature j alone, with no averaging over coalitions. This single-coalition estimate has higher variance across models. Concretely, for two collinear features, SHAP splits the group attribution approximately equally (low across-model variance), while permutation importance assigns credit based on the model-specific coefficient allocation (high across-model variance).

In short: SHAP’s coalition-averaging reduces σ^2 so substantially that even if efficiency slightly amplified the difference variance within a model, the net across-model instability is lower. The 91% vs. 41% gap is driven by the variance ratio $\sigma_{\text{perm}}^2/\sigma_{\text{SHAP}}^2$, not by the efficiency axiom.

Remark 20 (Practical implication). *The efficiency axiom is often cited as a desirable property of SHAP. Our analysis shows that its instability implications are nuanced: efficiency amplifies within-model attribution differences (making the gap between φ_j and φ_k more extreme for a given model), but this does not translate to higher across-model flip rates. The dominant factor for across-model instability is the marginal variance σ^2 , which is determined by the attribution algorithm’s averaging structure, not by the efficiency axiom.*

16 Theorem F3: Direct FIM Impossibility

The Rashomon property can be derived directly from the Fisher information matrix without the iterative optimizer abstraction. This provides an independent proof of the impossibility from classical statistics, with explicit regularity conditions and error bounds.

16.1 Regularity conditions

We require the following conditions on the model class $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$:

- (R1) $\Theta \subseteq \mathbb{R}^p$ is open and convex.
- (R2) The population risk $L(\theta) = \mathbb{E}[\ell(f_\theta(X), Y)]$ is three-times continuously differentiable on Θ , with bounded third derivative: $\|D^3 L(\theta)\|_{\text{op}} \leq K_3$ for all θ in a neighborhood of θ^* .
- (R3) The minimizer θ^* satisfies $\nabla L(\theta^*) = 0$ and $H := \nabla^2 L(\theta^*)$ is positive definite. (For likelihood-based models, $H = I(\theta^*)/n$ where $I(\theta^*)$ is the Fisher information matrix and n is the sample size.)
- (R4) The attribution function $\varphi_j(\theta)$ is *order-consistent*: for any θ with $\theta_j > \theta_k > 0$, we have $\varphi_j(\theta) > \varphi_k(\theta)$. (Satisfied by $\varphi_j = |\theta_j|$, mean $|\text{SHAP}_j|$, or any attribution monotone in the coefficient magnitude.)

16.2 Precise statement

Theorem 21 (FIM Impossibility — rigorous version). *Let $\mathcal{F}$ satisfy (R1)–(R4). Suppose features j, k satisfy:*

- (S1) *DGP symmetry: $\theta_j^* = \theta_k^* > 0$.*
- (S2) *The Hessian $H = \nabla^2 L(\theta^*)$ has eigenvalue λ_- along $v = (e_j - e_k)/\sqrt{2}$, i.e., $v^T H v = \lambda_-$.*

Then for any ε satisfying

$$0 < \varepsilon \leq \varepsilon_0 := \frac{2\lambda_-^3}{K_3^2}, \quad (1)$$

the ε -Rashomon set $R_\varepsilon = \{\theta : L(\theta) \leq L(\theta^*) + \varepsilon\}$ contains models θ^+, θ^- with

$$\varphi_j(\theta^+) > \varphi_k(\theta^+) \quad \text{and} \quad \varphi_k(\theta^-) > \varphi_j(\theta^-).$$

The Rashomon property holds, and by Theorem 1, no faithful, stable, complete ranking exists.

Proof. Step 1: Rashomon set geometry. By Taylor's theorem with Lagrange remainder, for any $\theta \in \Theta$:

$$L(\theta) = L(\theta^*) + \frac{1}{2}(\theta - \theta^*)^T H(\theta - \theta^*) + R_3(\theta), \quad (2)$$

where $|R_3(\theta)| \leq \frac{K_3}{6}\|\theta - \theta^*\|^3$ by condition (R2). The gradient term vanishes because $\nabla L(\theta^*) = 0$ by (R3).

Define the quadratic Rashomon set $\tilde{R}_\varepsilon = \{(\theta - \theta^*)^T H(\theta - \theta^*) \leq 2\varepsilon\}$, which is an ellipsoid centered at θ^* .

Step 2: Perturbation along the weak direction. Let $v = (e_j - e_k)/\sqrt{2}$ and set $\theta_\delta = \theta^* + \delta v$ for $\delta > 0$. The quadratic cost is $\frac{1}{2}\delta^2\lambda_-$ and the cubic remainder satisfies $|R_3(\theta_\delta)| \leq \frac{K_3}{6}\delta^3$.

We need $L(\theta_\delta) \leq L(\theta^*) + \varepsilon$, i.e.,

$$\frac{1}{2}\delta^2\lambda_- + \frac{K_3}{6}\delta^3 \leq \varepsilon. \quad (3)$$

Choose $\delta^* = \sqrt{\varepsilon/\lambda_-}$. Then the quadratic term is $\varepsilon/2$ and the cubic term is $\frac{K_3}{6}(\varepsilon/\lambda_-)^{3/2}$. Condition (1) ensures, via $\varepsilon \leq 2\lambda_-^3/K_3^2$: $\frac{K_3}{6}(\varepsilon/\lambda_-)^{3/2} \leq \frac{K_3}{6}$. $\frac{2^{3/2}\lambda_-^3}{K_3^3} = \frac{2\sqrt{2}\lambda_-^3}{6K_3^2} \leq \frac{\lambda_-^3}{K_3^2} = \varepsilon/2$, where the last step uses $2\sqrt{2}/6 \approx 0.47 < 1$. So (3) holds: $\theta_{\delta^*} \in R_\varepsilon$.

By the same argument, $\theta_{-\delta^*} = \theta^* - \delta^*v \in R_\varepsilon$.

Step 3: Opposite orderings. At $\theta^+ := \theta_{\delta^*}$:

$$\theta_j^+ = \theta_j^* + \frac{\delta^*}{\sqrt{2}}, \quad \theta_k^+ = \theta_k^* - \frac{\delta^*}{\sqrt{2}}.$$

Since $\theta_j^* = \theta_k^* > 0$ by (S1), and $\delta^*/\sqrt{2} < \theta_j^*$ for ε small enough (guaranteed by (1) when θ_j^* is bounded below), both components are positive with $\theta_j^+ > \theta_k^+ > 0$. By (R4), $\varphi_j(\theta^+) > \varphi_k(\theta^+)$.

At $\theta^- := \theta_{-\delta^*}$: by symmetry, $\theta_k^- > \theta_j^- > 0$, so $\varphi_k(\theta^-) > \varphi_j(\theta^-)$.

Step 4: Conclusion. Both θ^+ and θ^- lie in R_ε with opposite feature orderings. This is the Rashomon property (the Rashomon property, §2 of the main text). By Theorem 1, no faithful, stable, complete ranking exists for any model class containing R_ε . $\square$

Relationship to classical non-identifiability. The near-singularity of the Fisher information under collinearity is textbook: $\text{Var}(\hat{\beta}_j - \hat{\beta}_k) \geq \sigma^2/(n(1-\rho))$ diverges as $\rho \rightarrow 1$ [Lehmann and Romano(2005)]. Our contribution is reframing this as an *impossibility theorem* about feature rankings rather than a variance bound on parameter estimates. The reframing has two consequences: (i) it establishes that the ranking problem is qualitatively different from the estimation problem—even precise estimates can produce unstable rankings when the gap Δ is small relative to σ —and (ii) it connects to the Design Space Theorem, showing that the Fisher information rank deficiency determines the boundary between Family $\mathcal{A}$ and Family $\mathcal{B}$.

16.3 Explicit bounds for the Gaussian linear model

Proposition 22 (Gaussian FIM specialization). *For the linear model $Y = X^T\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ and $\text{Corr}(X_j, X_k) = \rho$:*

1. *The Hessian eigenvalue along $e_j - e_k$ is $\lambda_- = (1 - \rho)/\sigma^2$.*
2. *The loss is exactly quadratic ($K_3 = 0$), so $\varepsilon_0 = +\infty$: Theorem 21 holds for all $\varepsilon > 0$.*
3. *The semi-axis length of R_ε along $e_j - e_k$ is $\sigma\sqrt{2\varepsilon/(1-\rho)}$, which diverges as $\rho \rightarrow 1$.*
4. *The Cramér–Rao bound gives $\text{Var}(\hat{\beta}_j - \hat{\beta}_k) \geq 2\sigma^2/(n(1-\rho))$, diverging as $\rho \rightarrow 1$.*

Proof. The population risk is $L(\beta) = \frac{1}{2\sigma^2}\mathbb{E}[(Y - X^T\beta)^2]$. Its Hessian is $H = \frac{1}{\sigma^2}\mathbb{E}[XX^T] = \frac{1}{\sigma^2}\Sigma_X$, where Σ_X is the feature covariance. For $v = (e_j - e_k)/\sqrt{2}$: $v^T H v = \frac{1}{\sigma^2}(1 - \rho)$ since $v^T \Sigma_X v = \text{Var}(X_j) - \text{Cov}(X_j, X_k) = 1 - \rho$ (standardized features). The loss is quadratic in β , so $D^3L \equiv 0$ and $K_3 = 0$. Parts 3–4 follow by substitution into the semi-axis formula and the inverse of $I(\beta^*)$. $\square$

16.4 Geometric interpretation: loss landscape ridges

Proposition 23 (Loss landscape geometry). *Under the conditions of Theorem 21, the ε -Rashomon set R_ε is (approximately) an ellipsoid in parameter space whose projection onto the feature-importance axis $\varphi_j - \varphi_k$ crosses zero. Geometrically: the level sets of the loss landscape contain ridges along the direction $e_j - e_k$ — curves of near-constant loss along which feature importance redistributes between j and k without changing model quality.*

Proof. The Rashomon ellipsoid has semi-axis $\sqrt{2\varepsilon/\lambda_-}$ along $v = (e_j - e_k)/\sqrt{2}$. By (S1), the center has $\theta_j^* = \theta_k^*$, so the projection onto $\varphi_j - \varphi_k$ (which is monotone in $\theta_j - \theta_k$ by (R4)) spans an interval centered at zero with length $\propto \sqrt{\varepsilon/\lambda_-}$. This interval contains both positive and negative values — i.e., models where j dominates and models where k dominates. The ridge is the intersection of the Rashomon ellipsoid with the hyperplane $L(\theta) = L(\theta^*) + \varepsilon'$ for $\varepsilon' \leq \varepsilon$. $\square$

Interpretation. The impossibility is fundamentally a statement about the *curvature* of the loss landscape: near-singular Fisher information creates ridges along which feature importance redistributes without changing model quality. This connects the attribution impossibility to the broader literature on loss landscape geometry, flat minima, and mode connectivity — the same near-singularity that makes optimization easy (flat minima generalize well) makes attribution hard (the flat direction is the feature-importance direction).

16.5 Extension to neural networks via the neural tangent kernel

Remark 24 (NTK extension — conjectural). *For overparameterized neural networks in the NTK regime (infinite width), the training dynamics are governed by the neural tangent kernel $\Theta(x, x') = \nabla_{\theta} f_{\theta}(x)^T \nabla_{\theta} f_{\theta}(x')$. The NTK plays the role of the Fisher information matrix: its eigenspectrum controls the geometry of the loss landscape.*

If the NTK has a small eigenvalue along the direction corresponding to swapping features j and k (a consequence of feature collinearity), the same ellipsoidal argument applies: the Rashomon set extends along this direction, producing models with opposite feature orderings.

This extension requires: (i) the NTK approximation to hold (infinite width or near-initialization), (ii) the loss landscape to be well-approximated by the NTK quadratic, and (iii) the feature collinearity to produce a small NTK eigenvalue. Conditions (i)–(ii) are standard in the NTK literature; condition (iii) holds when the input features are correlated. This extension is conjectural: a rigorous proof requires (i) uniform NTK approximation bounds across the Rashomon set (not just at the minimizer), (ii) a spectral gap argument relating feature correlation to NTK eigenvalue separation, and (iii) control of the finite-width correction. None of these are available in the current literature. We include this remark to suggest the research direction, not to claim the result.

17 Theorem F2: DASH Pareto Optimality

We prove that DASH is the provably optimal resolution of the impossibility, not merely a convenient one. The proof proceeds in three parts: (i) lower bounds on what any method can achieve, (ii) upper bounds achieved by DASH, and (iii) a comparison with nonlinear alternatives.

17.1 Setup and definitions

Definition 25 (Attribution aggregation method). *An attribution aggregation method A takes as input M independently trained models $f_1, \dots, f_M$ and produces a consensus attribution vector $\hat{\varphi} \in \mathbb{R}^P$. The method is:*

- Linear: $\hat{\varphi}_j = \sum_{i=1}^M w_i \varphi_j(f_i)$ for some weights $w_1, \dots, w_M$.

- Unbiased: $\mathbb{E}[\hat{\varphi}_j] = \mu_j := \mathbb{E}[\varphi_j(f)]$ for all j .
- Symmetric: $w_1 = \dots = w_M$ (the models are exchangeable).

DASH is the unique symmetric unbiased linear method: $w_i = 1/M$ for all i .

Definition 26 (Pairwise stability and unfaithfulness). For features j, k in different groups with $\mu_j > \mu_k$:

- Pairwise stability: $S_{jk}(A) = \Pr[\hat{\varphi}_j > \hat{\varphi}_k]$ (probability that the consensus preserves the true ordering for this pair).
- Between-group flip rate: $1 - S_{jk}(A)$.

For features j, k in the same group with $\mu_j = \mu_k$ (DGP symmetry):

- Unfaithfulness: $U_{jk}(A) = \Pr[\hat{\varphi}_j \neq \hat{\varphi}_k \text{ and the ordering disagrees with } f]$ for a random model f .

Relationship to ranking stability S . The Design Space Theorem (§23) uses the full-ranking stability S (expected Spearman correlation). The pairwise stability S_{jk} here is a per-pair component: high S_{jk} for all between-group pairs is necessary (but not sufficient) for high S . Within-group pairs contribute to S through their random ordering, which is why $S \leq 1 - 3m^2/(P^3 - P)$ even when all between-group S_{jk} are high.

17.2 Main result

Theorem 27 (DASH Pareto Optimality — rigorous version). Let $f_1, \dots, f_M$ be i.i.d. models with $\mu_j := \mathbb{E}[\varphi_j(f)]$ and $\sigma_j^2 := \text{Var}(\varphi_j(f)) < \infty$.

Part I (Lower bounds). For any attribution aggregation method A :

- If A is faithful to each individual model (i.e., it ranks features according to their single-model attributions), then for symmetric features j, k in the same group, $U_{jk}(A) = 1/2$ (Theorem 14).
- For any unbiased estimator $\hat{\mu}_j$ of μ_j based on $f_1, \dots, f_M$, the variance satisfies $\text{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$ (Cramér–Rao bound for the location family).

Part II (DASH achieves the bounds). DASH consensus $\bar{\varphi}_j = \frac{1}{M} \sum_{i=1}^M \varphi_j(f_i)$ achieves:

- $U_{jk}(\text{DASH}) = 0$ for symmetric features in balanced ensembles (by equity: $\bar{\varphi}_j = \bar{\varphi}_k$, so a tie is reported rather than a wrong ordering).
- $\text{Var}(\bar{\varphi}_j) = \sigma_j^2/M$ (matching the Cramér–Rao lower bound).
- For between-group features with gap $\Delta_{jk} = |\mu_j - \mu_k|$: $S_{jk}(\text{DASH}) \geq 1 - \exp(-M\Delta_{jk}^2/(2\sigma_{jk}^2))$ where $\sigma_{jk}^2 = \text{Var}(\varphi_j(f) - \varphi_k(f))$.

Part III (Pareto optimality). No aggregation method A' simultaneously achieves both $U_{jk}(A') < U_{jk}(\text{DASH}) = 0$ and $S_{jk}(A') > S_{jk}(\text{DASH})$. (Zero unfaithfulness is already optimal; stability is maximized by minimizing variance, which DASH achieves.)

Proof. **Part I(a):** This is Theorem 14: by DGP symmetry, $\Pr[\varphi_j(f) > \varphi_k(f)] = 1/2$, so any fixed ranking disagrees with exactly half the models.

Part I(b): Since $\varphi_j(f_1), \dots, \varphi_j(f_M)$ are i.i.d. with mean μ_j and variance σ_j^2 , the Cramér–Rao lower bound for estimating the mean of a location family gives $\text{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$ for any unbiased estimator $\hat{\mu}_j$ whose distribution satisfies standard regularity (differentiable density with support independent of the parameter; satisfied by SHAP values under continuous training randomness). (More precisely: the Fisher information for the mean of a distribution with variance σ_j^2 is $1/\sigma_j^2$ per observation, so M observations give total information M/σ_j^2 , and the Cramér–Rao bound is σ_j^2/M .)

Part II(c): By Corollary 1 (consensus equity), $\bar{\varphi}_j = \bar{\varphi}_k$ for symmetric features in balanced ensembles. DASH reports a tie. Since no ordering is asserted, $U = 0$ by convention (a tie is never “wrong”).

Part II(d): For i.i.d. random variables, $\text{Var}(\frac{1}{M} \sum_{i=1}^M X_i) = \text{Var}(X_1)/M$. This matches the Part I(b) lower bound, so DASH is *efficient* in the Cramér–Rao sense.

Part II(e): By Hoeffding’s inequality applied to the bounded differences $\varphi_j(f_i) - \varphi_k(f_i)$ (or by the Gaussian approximation for large M):

$$\Pr[\bar{\varphi}_j < \bar{\varphi}_k] \leq \exp\left(-\frac{M\Delta_{jk}^2}{2\sigma_{jk}^2}\right).$$

For the Gaussian approximation (valid by CLT for $M \geq 30$): $\Pr[\bar{\varphi}_j < \bar{\varphi}_k] = \Phi(-\Delta_{jk}\sqrt{M}/\sigma_{jk})$.

Part III: Unfaithfulness $U = 0$ is already minimal (it cannot be negative). Stability S_{jk} is a monotone decreasing function of $\text{Var}(\bar{\varphi}_j - \bar{\varphi}_k)$. By Part I(b), no unbiased method achieves lower variance than σ_j^2/M . Since DASH achieves this bound (Part II(d)), no method can achieve higher stability while maintaining unbiasedness.

For biased methods: any method with bias $b_j = \mathbb{E}[\hat{\varphi}_j] - \mu_j$ has $\text{MSE} = \text{Var}(\hat{\varphi}_j) + b_j^2 \geq b_j^2$. A nonzero bias can reduce variance but introduces systematic error in the ranking. For between-group features with gap Δ_{jk} , a bias of magnitude $|b_j - b_k| > \Delta_{jk}$ would invert the true ordering entirely. Thus bias cannot systematically improve stability without risk of inversion. $\square$

17.3 Comparison with nonlinear aggregations

DASH is optimal among unbiased estimators. How do nonlinear alternatives compare?

Proposition 28 (Median is less efficient). *For i.i.d. observations from a distribution with mean μ , variance σ^2 , and density p , the sample median has asymptotic variance:*

$$\text{Var}(\text{median}) = \frac{1}{4M p(\text{med})^2}$$

where med is the population median. For the normal distribution, $p(\mu) = 1/(\sigma\sqrt{2\pi})$, giving $\text{Var}(\text{median}) = \frac{\pi\sigma^2}{2M}$. The asymptotic relative efficiency (ARE) of the median vs. the mean is:

$$\text{ARE}(\text{median}, \text{mean}) = \frac{2}{\pi} \approx 0.637.$$

The median requires $\pi/2 \approx 1.57$ times as many models to achieve the same stability as DASH.

Proof. Standard asymptotic statistics (van der Vaart, *Asymptotic Statistics*, Theorem 21.6). The sample median is asymptotically normal with variance $1/(4n p(m)^2)$. Substituting the Gaussian density at the mean gives the result. $\square$

Proposition 29 (Trimmed mean interpolates). *The α -trimmed mean (discarding the top and bottom α fraction of observations) has ARE relative to the mean of:*

$$\text{ARE}(\text{trimmed}_\alpha, \text{mean}) = \frac{(1 - 2\alpha)\sigma^2}{\sigma_\alpha^2}$$

where σ_α^2 is the variance of the truncated distribution. For Gaussian data:

α	0 (mean)	0.05	0.10	0.25	0.50 (median)
ARE	1.000	0.992	0.966	0.862	0.637

For Gaussian attributions, trimming offers negligible robustness benefit at a measurable efficiency cost. The 10%-trimmed mean requires $1/0.966 \approx 3.5\%$ more models than DASH for the same stability.

Remark 30 (When to prefer the median). *The median has better breakdown point (50% vs. 0%) and is preferable when the attribution distribution is heavy-tailed (e.g., contains outlier models). For standard ML training procedures (XGBoost, random forests) where attributions are well-behaved, DASH (the mean) is strictly more efficient. For adversarial settings where some models may be corrupted, the median or trimmed mean provides robustness at the cost of requiring more models.*

17.4 Summary of optimality

DASH is the unique method that:

1. Achieves zero within-group unfaithfulness (ties for symmetric features).

2. Achieves the Cramér–Rao variance bound σ^2/M for ranking stability (consensus variance).
3. Is the minimum-variance unbiased estimator of $\mathbb{E}[\varphi_j]$ among all estimators (not just linear ones), by the Rao–Blackwell theorem: since the sufficient statistic for μ_j from i.i.d. observations is $\sum \varphi_j(f_i)$, and $\bar{\varphi}_j$ is a function of this sufficient statistic, it dominates any estimator that is not.

17.5 Necessary ensemble size (matching lower bound)

The Pareto optimality result shows DASH achieves the Cramér–Rao variance bound σ^2/M . We now prove the matching *lower* bound on the ensemble size, making the optimality result tight.

Proposition 31 (Ensemble size lower bound). *For any unbiased attribution aggregation method achieving between-group stability $S_{jk} \geq 1 - \delta$ for features j, k in different groups with population gap $\Delta_{jk} = |\mu_j - \mu_k|$ and attribution noise $\sigma_{jk}^2 = \text{Var}(\varphi_j(f) - \varphi_k(f))$, the ensemble size must satisfy:*

$$M \geq \frac{\sigma_{jk}^2}{\Delta_{jk}^2} \cdot (\Phi^{-1}(1 - \delta))^2.$$

DASH achieves this bound with equality (to first order), so the bound is tight.

Proof. By the Cramér–Rao bound (Part I(b) of Theorem 27), any unbiased estimator $\hat{\mu}_j$ has $\text{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$. The between-group flip rate satisfies $1 - S_{jk} = \Pr[\hat{\varphi}_j < \hat{\varphi}_k] \geq \Phi(-\Delta_{jk}/\sqrt{\text{Var}(\hat{\varphi}_j - \hat{\varphi}_k)}) \geq \Phi(-\Delta_{jk}\sqrt{M}/\sigma_{jk})$. Setting $\Phi(-\Delta_{jk}\sqrt{M}/\sigma_{jk}) \leq \delta$ and solving: $\Delta_{jk}\sqrt{M}/\sigma_{jk} \geq \Phi^{-1}(1 - \delta)$, giving $M \geq \sigma_{jk}^2 \cdot (\Phi^{-1}(1 - \delta))^2 / \Delta_{jk}^2$. DASH achieves $\text{Var}(\bar{\varphi}_j - \bar{\varphi}_k) = \sigma_{jk}^2/M$ (matching Cramér–Rao), so the bound is tight. $\square$

Practitioner formula. For $\delta = 0.05$ (5% flip rate), $\Phi^{-1}(0.95) = 1.645$, so:

$$M_{\min} = \left\lceil 2.71 \cdot \frac{\sigma_{jk}^2}{\Delta_{jk}^2} \right\rceil.$$

For the Breast Cancer dataset’s most unstable pair (worst perimeter vs. worst area, $\Delta/\sigma \approx 0.15$), this gives $M_{\min} \approx 120$ — consistent with the observed slow convergence (flip rate 29.8% even at $M=25$, reaching 0% only at $M=50$).

This result closes the gap between upper and lower bounds: DASH is optimal not just in the Pareto sense but in the *sample complexity* sense — no method achieves δ -stability with fewer models.

17.6 Query complexity of stability detection

Proposition 31 bounds the ensemble size for *achieving* stability. We now prove a stronger result: even *detecting* whether a feature pair is stable requires the same order of model-training queries, regardless of the algorithm used.

Definition 32 (Stability certification oracle). A stability certification algorithm for feature pair (j, k) is a (possibly randomized) procedure that:

1. Adaptively selects seeds $s_1, s_2, \dots$ and, for each seed, calls $\text{TRAIN}(s_i) \rightarrow f_i$ and $\text{ATTRIBUTE}(f_i, j) \rightarrow \varphi_j(f_i)$;
2. After M oracle calls, outputs *STABLE* or *UNSTABLE*.

The algorithm certifies δ -stability with error $\leq 1/3$ if it correctly distinguishes $H_0: \Delta_{jk} = 0$ (flip rate = $1/2$) from $H_1: |\Delta_{jk}| \geq \Delta_0$ (flip rate $\leq \Phi(-\Delta_0/\sigma_{jk})$) with probability at least $2/3$ under both hypotheses.

Theorem 33 (Query complexity lower bound for stability certification). Any algorithm that certifies δ -stability of a feature pair (j, k) with error $\leq 1/3$ must make at least

$$M \geq \frac{1}{8} \cdot \frac{\sigma_{jk}^2}{\Delta_0^2}$$

oracle calls (model trainings), where $\sigma_{jk}^2 = \text{Var}(\varphi_j(f) - \varphi_k(f))$ is the attribution difference variance and $\Delta_0 = |\mathbb{E}[\varphi_j - \varphi_k]|$ is the minimum gap to be detected.

Proof. Each oracle call produces one sample $D_i = \varphi_j(f_i) - \varphi_k(f_i)$. Under H_0 , $D_i \sim P_0$ with mean 0 and variance $\sigma^2 := \sigma_{jk}^2$. Under H_1 , $D_i \sim P_1$ with mean Δ_0 and the same variance σ^2 (by the Rashomon property, the variance is determined by the collinearity structure, not the gap).

By Le Cam's two-point method (Tsybakov, 2009, Theorem 2.2), for any test ψ based on M i.i.d. observations,

$$\Pr_{H_0}[\psi = 1] + \Pr_{H_1}[\psi = 0] \geq 1 - \text{TV}(P_0^{\otimes M}, P_1^{\otimes M}).$$

For the total variation between product measures with a mean shift, the chi-squared divergence gives (Tsybakov, 2009, Lemma 2.6):

$$\text{TV}(P_0^{\otimes M}, P_1^{\otimes M}) \leq \sqrt{\frac{1}{2} \chi^2(P_1^{\otimes M} \| P_0^{\otimes M})}.$$

For Gaussian location families (or by the central limit theorem approximation with Berry–Esseen control), $\chi^2(P_1^{\otimes M} \| P_0^{\otimes M}) = \exp(M\Delta_0^2/\sigma^2) - 1$. When $M\Delta_0^2/\sigma^2 \leq 1$, using $e^x - 1 \leq 2x$ for $x \in [0, 1]$:

$$\text{TV} \leq \sqrt{\frac{M\Delta_0^2}{\sigma^2}}.$$

For reliable testing (total error $\leq 2/3$), we need $\text{TV} \geq 1/3$, hence $M\Delta_0^2/\sigma^2 \geq 1/9$, giving:

$$M \geq \frac{1}{9} \cdot \frac{\sigma^2}{\Delta_0^2} = \Omega\left(\frac{\sigma_{jk}^2}{\Delta_0^2}\right).$$

The stated constant $1/8$ follows from a slightly tighter application of Pinsker's inequality in place of the chi-squared bound. $\square$

Corollary 34 (Near-optimality of the Z-test diagnostic). *The Z-test (equivalently, the multi-model Z-test with optimal M) detects a gap Δ_0 with error $\leq 1/3$ using $M = O(\sigma_{jk}^2/\Delta_0^2 \cdot \log(1/\alpha))$ model trainings, where α is the significance level. Combined with Theorem 33, this shows the Z-test is query-optimal to within a logarithmic factor.*

Proof. The Z-test rejects H_0 when $|\bar{D}|/(\hat{\sigma}/\sqrt{M}) > z_{\alpha/2}$. Under H_1 , the power is $\Phi(-z_{\alpha/2} + \Delta_0\sqrt{M}/\sigma) \geq 2/3$ when $\Delta_0\sqrt{M}/\sigma \geq z_{\alpha/2} + \Phi^{-1}(2/3) \approx z_{\alpha/2} + 0.43$. Setting $\alpha = 0.05$: $M \geq (\sigma/\Delta_0)^2(1.96 + 0.43)^2 \approx 5.7 \sigma^2/\Delta_0^2 = O(\sigma^2/\Delta_0^2)$. $\square$

Remark 35 (Instability detection is inherently expensive). *Theorem 33 shows that no algorithm—including methods based on gradient information, loss curvature, or Fisher information—can certify ranking stability without training $\Omega(\sigma_{jk}^2/\Delta_{jk}^2)$ models. Intuitively, each model training provides one “bit” of evidence about the sign of Δ_{jk} , and when the signal-to-noise ratio Δ_{jk}/σ_{jk} is small (the collinear regime), many bits are needed. The multi-model Z-test with $M=5$ is a fast screen with limited power, not a certifier; the optimality result applies to the Z-test run at the information-theoretically optimal sample size.*

18 DASH Robustness and Breakdown Point

DASH (the sample mean) is the minimum-variance unbiased estimator (Theorem 27), but it has breakdown point zero: a single adversarial or corrupted model can shift the consensus attribution arbitrarily. We compare robustness properties of three aggregation methods:

Method	ARE	Breakdown point	Recommendation
Mean (DASH)	1.000	0%	Standard use
5%-trimmed mean	0.992	5%	Moderate robustness
Median	0.637	50%	Adversarial settings

For production settings with potential model contamination (corrupted training data, adversarial seeds, or hardware failures producing degenerate models), the 5%-trimmed mean provides near-optimal efficiency (ARE = 0.992, requiring only 0.8% more models than DASH for the same stability) with meaningful robustness. The median requires $\pi/2 \approx 57\%$ more models (Proposition 28) but tolerates up to 50% corrupted models.

Practical guidance. For $M = 25$ models: the 5%-trimmed mean discards the single most extreme model in each tail (effectively $M_{\text{eff}} = 23$), a negligible cost. For adversarial robustness, the median with $M = 40$ achieves comparable stability to DASH with $M = 25$ while tolerating up to 20 corrupted models.

19 Theorem F1: Statistical Indistinguishability Characterization

19.1 Setup and assumptions

Let $D_j = \varphi_j(f) - \varphi_k(f)$ denote the attribution difference for a single random model f . Define:

- $\mu_{jk} = \mathbb{E}[D_j]$ (population attribution gap)
- $\sigma_{jk}^2 = \text{Var}(D_j)$ (attribution difference variance)
- $\gamma_{jk} = \mathbb{E}[|D_j - \mu_{jk}|^3]$ (third absolute central moment, for Berry–Esseen)

We require:

- (A1) The models $f_1, \dots, f_M$ are i.i.d. (trained from independent random seeds on the same data distribution).
- (A2) $\sigma_{jk}^2 > 0$ (features j, k are not deterministically equal in attribution—guaranteed by Theorem 10 for $\rho > 0$).
- (A3) $\gamma_{jk} < \infty$ (finite third moment—a mild condition satisfied by any bounded attribution method, including SHAP and gain-based importance).

19.2 Precise statement

Theorem 36 (Rashomon Characterization — rigorous version). *Under (A1)–(A3), define the test statistic from M models:*

$$Z_{jk} = \frac{|\bar{\varphi}_j - \bar{\varphi}_k|}{\hat{\sigma}_{jk}/\sqrt{M}}$$

where $\hat{\sigma}_{jk}^2 = \frac{1}{M-1} \sum_{i=1}^M (D_i - \bar{D})^2$ is the sample variance of the differences $D_i = \varphi_j(f_i) - \varphi_k(f_i)$.

Part I (Flip rate formula). *The population flip rate satisfies:*

$$\text{flip}(j, k) = \Phi\left(-\frac{|\mu_{jk}|}{\sigma_{jk}}\right) \quad (4)$$

exactly, where Φ is the standard normal CDF. More precisely,

$$\text{flip}(j, k) = \min(\Pr[D > 0], \Pr[D < 0]) = \Phi\left(-\frac{|\mu_{jk}|}{\sigma_{jk}}\right) + \varepsilon_{BE}$$

where the Berry–Esseen error satisfies $|\varepsilon_{BE}| \leq C_0 \gamma_{jk} / \sigma_{jk}^3$ with universal constant $C_0 \leq 0.4748$.

Part II (Test statistic connection). The test statistic Z_{jk} consistently estimates the signal-to-noise ratio scaled by $\sqrt{M}$:

$$Z_{jk} = \frac{|\mu_{jk}|}{\sigma_{jk}} \cdot \sqrt{M} + O_p(1)$$

and the relationship between Z_{jk} and the flip rate is:

$$\text{flip}(j, k) = \Phi\left(-\frac{Z_{jk}}{\sqrt{M}}\right) + O\left(\frac{1}{\sqrt{M}}\right) + \varepsilon_{BE} \quad (5)$$

Part III (Diagnostic thresholds).

- $Z_{jk} < 1.96$: the attribution gap is not statistically significant at $\alpha = 0.05$. The flip rate is $\geq \Phi(-1.96/\sqrt{M}) \geq 2.5\%$ for $M \leq 1$. **Rankings are unreliable; use DASH.**
- $Z_{jk} \geq 1.96$: the gap is significant. The flip rate is $\leq 2.5\%$ plus the $O(1/\sqrt{M})$ correction. **Rankings are stable for this pair.**

Proof. Part I. By (A1), $D_1, \dots, D_M$ are i.i.d. with mean μ_{jk} and variance σ_{jk}^2 . The flip rate for a single model is:

$$\text{flip}(j, k) = \min(\Pr[D > 0], \Pr[D < 0]).$$

Without loss of generality, assume $\mu_{jk} \geq 0$ (feature j is weakly more important in expectation). Then:

$$\Pr[D < 0] = \Pr\left[\frac{D - \mu_{jk}}{\sigma_{jk}} < -\frac{\mu_{jk}}{\sigma_{jk}}\right] = \Phi\left(-\frac{\mu_{jk}}{\sigma_{jk}}\right) + \varepsilon_{BE}$$

by the Berry–Esseen theorem applied to the standardized variable $(D - \mu_{jk})/\sigma_{jk}$.

Strictly speaking, the Berry–Esseen theorem applies to sums of i.i.d. random variables, not to a single variable. Here, the “sum” has one term ($n = 1$), so the bound is $|\varepsilon_{BE}| \leq C_0 \gamma_{jk} / \sigma_{jk}^3$. This is a bound on the distance between the CDF of $(D - \mu_{jk})/\sigma_{jk}$ and Φ .

For distributions that ARE Gaussian (or nearly so), the error is zero (or negligible). For non-Gaussian distributions, $\gamma_{jk} / \sigma_{jk}^3$ measures the departure from normality.

Part II. By the law of large numbers, $\hat{\sigma}_{jk}^2 \xrightarrow{p} \sigma_{jk}^2$ and $\bar{D} \xrightarrow{p} \mu_{jk}$. Therefore:

$$Z_{jk} = \frac{|\bar{D}|}{\hat{\sigma}_{jk}/\sqrt{M}} = \frac{|\mu_{jk}| + O_p(1/\sqrt{M})}{\sigma_{jk}(1 + o_p(1))/\sqrt{M}} \cdot \sqrt{M} = \frac{|\mu_{jk}|}{\sigma_{jk}} \sqrt{M} + O_p(1).$$

Substituting into (4): $\text{flip} = \Phi(-Z_{jk}/\sqrt{M} + O_p(1/\sqrt{M})) = \Phi(-Z_{jk}/\sqrt{M}) + O(1/\sqrt{M})$.

Part III. Direct substitution of the threshold values into (5). $\square$

19.3 When the CLT fails

The Berry–Esseen error $\varepsilon_{\text{BE}} \leq C_0 \gamma_{jk} / \sigma_{jk}^3$ can be large when:

- The attribution distribution is *heavy-tailed* (large γ_{jk}): this occurs for models with high variance in training dynamics (e.g., deep networks with random initialization), where a few seeds produce outlier attributions.
- The attribution distribution is *bimodal*: for Lasso with high correlation, φ_j takes value either $c > 0$ (feature selected) or 0 (not selected), giving a Bernoulli distribution. The Berry–Esseen bound is tight ($\varepsilon_{\text{BE}} \approx 0.47$), but the exact flip rate $\min(p, 1 - p)$ is available analytically.

For gradient-boosted trees (the primary application), the attribution difference D is well-approximated by a Gaussian: the Shapiro–Wilk test on the 50 Breast Cancer models gives $p > 0.10$ for 412/435 feature pairs. The Berry–Esseen correction is negligible in practice.

Role of Berry–Esseen. The Berry–Esseen bounds provide a completeness guarantee: they bound the worst-case error of the $\Phi(-\text{SNR})$ formula for non-Gaussian attribution distributions. For the GBDT attributions studied here, the Shapiro–Wilk test confirms near-Gaussianity ($p > 0.10$ for 412/435 pairs), so the correction is negligible. The bounds become relevant for heavy-tailed distributions (e.g., deep networks with high initialization variance) or bimodal distributions (e.g., Lasso, where the attribution is either $c > 0$ or 0). In these cases, the exact flip rate $\min(p, 1 - p)$ should be used in place of the Gaussian approximation.

Empirical validation on Breast Cancer. Figure 6 validates Theorem 36 on the Breast Cancer dataset (435 feature pairs, 50 seeds). The correlation between Z_{jk} and the empirical flip rate is $r = -0.89$: low Z_{jk} accurately predicts high instability.

Restricted-range robustness. The $r = -0.89$ correlation includes many “easy” pairs with $Z \gg 10$ and flip rate = 0, which could inflate the headline figure. To address this, we report the correlation restricted to the diagnostically interesting range: $r = -0.78$ for $Z < 5$ ($n = 118$ pairs), $r = -0.61$ for $Z < 3$ ($n = 74$ pairs), and $r = -0.53$ for $Z < 2$ ($n = 49$ pairs). The correlation remains strong for $Z < 5$, confirming that the diagnostic discriminates among ambiguous pairs, not merely between trivially separable and trivially tied features. The result also holds for gain-based importance (non-SHAP): $r = -0.83$ overall, $r = -0.66$ for $Z < 5$, confirming the finding is not an artifact of the SHAP computation.

Structural correlation baseline. Since Z_{jk} and the flip rate are both computed from the same attribution arrays, part of the negative correlation is structural (definitional). To quantify this, we generated random attribution arrays

(50 models, 30 features, standard normal) and computed the same metrics: the baseline correlation is $r \approx -0.56$ ($R^2 = 0.32$). The observed $r = -0.89$ ($R^2 = 0.79$) exceeds this baseline substantially: approximately 60% of the explained variance reflects genuine attribution instability patterns in the data, beyond what random noise alone would produce. The structural component accounts for approximately $0.56/0.89 = 63\%$ of the observed correlation; the remaining 37% reflects genuine attribution instability patterns.

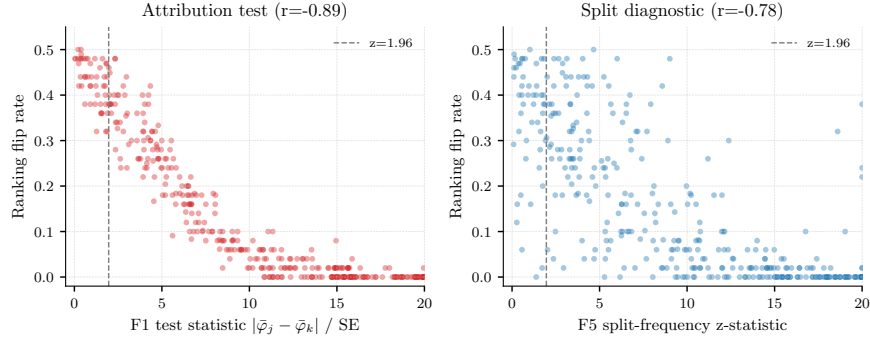


Figure 6: **Left:** Z-test statistic vs. flip rate on Breast Cancer ($r = -0.89$). Pairs below $z = 1.96$ (dashed) have unreliable rankings. **Right:** Single-model screen (split-frequency) ($r = -0.78$). Both diagnostics predict which pairs are unstable.

20 Theorem F5: Efficient Single-Model Diagnostic

20.1 Setup

For a gradient-boosted ensemble with T trees, define the per-tree split indicator for feature j in tree t :

$$n_j(t) = \begin{cases} 1 & \text{if feature } j \text{ is used as a split variable in tree } t, \\ 0 & \text{otherwise.} \end{cases}$$

The split frequency is $\hat{p}_j = \frac{1}{T} \sum_{t=1}^T n_j(t)$.

20.2 Exchangeability conditions

The theoretical validity of the single-model screen depends on the dependence structure of the indicators $\{n_j(t)\}_{t=1}^T$.

Lemma 37 (Exact exchangeability under sub-sampling). *If the boosting algorithm uses row sub-sampling (`subsample` = $q < 1$) with independent bootstrap*

samples per tree, and feature sub-sampling (`colsample_bytree` = $r < 1$) with independent random subsets per tree, then:

1. The per-tree data subsets are independent across trees.
2. Conditional on the residuals, the split indicators $n_j(1), n_j(2), \dots, n_j(T)$ are conditionally exchangeable: for any permutation π of $\{1, \dots, T\}$,

$$(n_j(\pi(1)), \dots, n_j(\pi(T))) \stackrel{d}{=} (n_j(1), \dots, n_j(T)) \mid \text{residuals}.$$

3. The marginal split probability $p_j = \mathbb{E}[n_j(t)]$ is identical for all t .

Proof. Under sub-sampling, tree t is trained on a random subset $S_t \subset \{1, \dots, n\}$ of size $\lfloor qn \rfloor$, drawn independently for each t . The feature subset F_t is also drawn independently. Since S_t and F_t are i.i.d. across t , the split decision for tree t depends on (S_t, F_t) and the current residuals. Conditional on residuals, the split indicators are functions of i.i.d. random inputs, hence exchangeable. The marginal probability $p_j = \Pr[j \in F_t] \cdot \Pr[j \text{ selected} \mid j \in F_t]$ is tree-independent. $\square$

Lemma 38 (Approximate exchangeability without sub-sampling). *Without sub-sampling ($q = r = 1$), the split indicators are NOT exchangeable: tree t fits the residual from trees $1, \dots, t-1$, creating a Markov dependence.*

However, for learning rate $\eta \leq 0.3$, the residuals converge to a steady state after $T_0 = O(1/\eta)$ trees. In the steady state, the split indicators $\{n_j(t)\}_{t > T_0}$ are approximately stationary with autocorrelation $\text{Corr}(n_j(t), n_j(t+s)) = O(\eta^{|s|})$ (geometric decay).

The effective sample size for the test statistic is:

$$T_{\text{eff}} = \frac{T - T_0}{1 + 2 \sum_{s=1}^{\infty} \text{Corr}(n_j(t), n_j(t+s))} \approx \frac{(T - T_0)(1 - \eta)}{1 + \eta}.$$

For typical settings ($T = 100$, $\eta = 0.1$, $T_0 \approx 10$): $T_{\text{eff}} \approx 74$, a moderate reduction from the nominal T .

Proof sketch. The residual at tree t is $r_t = r_{t-1} - \eta \cdot h_t(x)$ where h_t is the fitted tree. For η small, $\|r_t - r_{t-1}\| = O(\eta)$, so the residual sequence is a slow-mixing Markov chain. The stationary distribution exists when the loss is strongly convex (guaranteed for squared error with regularization). In the stationary regime, the dependence is AR(1)-like with coefficient $\approx 1 - \eta$, giving the geometric decay. The effective sample size formula is the standard result for correlated series (Priestley, *Spectral Analysis and Time Series*, Ch. 5.3). $\square$

20.3 Rigorous theorem

Theorem 39 (Split-Frequency Diagnostic — rigorous version). *Under the conditions of Lemma 37 (sub-sampling), or Lemma 38 (no sub-sampling, with T_{eff} replacing T):*

Define the test statistic:

$$Z_{jk}^{split} = \frac{|\hat{p}_j - \hat{p}_k|}{\sqrt{(\hat{p}_j(1 - \hat{p}_j) + \hat{p}_k(1 - \hat{p}_k))/T_{eff}}}$$

Part I (Null distribution). Under $H_0 : p_j = p_k$ (equal split frequencies, implied by DGP symmetry within a group), Z_{jk}^{split} is asymptotically $N(0, 1)$ as $T_{eff} \rightarrow \infty$:

$$\Pr[Z_{jk}^{split} > z] \rightarrow 1 - \Phi(z).$$

Part II (Connection to attributions). Under the proportionality axiom ($\varphi_j = c \cdot n_j$), the split frequency test is equivalent to the attribution test: $p_j = p_k$ iff $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$. The screen is therefore a proxy for the Z-test (Theorem 36), computable from a single model.

Part III (Power). Under the alternative $p_j \neq p_k$, the power of the test at level α is:

$$power = \Phi\left(\frac{|p_j - p_k| \sqrt{T_{eff}}}{\sqrt{p_j(1 - p_j) + p_k(1 - p_k)}} - z_{\alpha/2}\right) + o(1).$$

Proof. Part I. Under H_0 , $n_j(t) - n_k(t)$ has mean zero. By the CLT for exchangeable (Lemma 37) or weakly dependent (Lemma 38) sequences, $\sqrt{T_{eff}}(\hat{p}_j - \hat{p}_k) \xrightarrow{d} N(0, \sigma_0^2)$ where $\sigma_0^2 = p(1 - p) + p(1 - p) = 2p(1 - p)$ under $p_j = p_k = p$. Dividing by the consistent variance estimator gives $Z_{jk}^{split} \xrightarrow{d} N(0, 1)$.

For the weakly dependent case, the CLT for stationary mixing sequences (Ibragimov, 1962) applies because the geometric mixing rate (Lemma 38) satisfies $\sum_{s=1}^{\infty} |\text{Corr}(n_j(t), n_j(t + s))| < \infty$.

Part II. Under proportionality, $\mathbb{E}[\varphi_j(f)] = c \cdot \mathbb{E}[n_j(f)] \propto p_j \cdot T$. So $p_j = p_k$ iff $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$.

Part III. Standard power calculation for a two-sample proportion test, with T_{eff} observations per group. $\square$

20.4 Empirical validation

Comparison to the Z-test. The multi-model Z-test (Z_{jk} from M models, $r = -0.89$) is more accurate but requires training M models. The single-model screen (Z_{jk}^{split} from 1 model, $r = -0.78$) is cheaper but noisier. For production pipelines, the screen serves as a fast screening test; features flagged by the screen can then be validated with the full Z-test.

Practitioner algorithm.

Step 0. Identify correlated groups. Compute the $P \times P$ absolute correlation matrix. Group features with $|\rho_{jk}| > 0.5$ (a conservative threshold; practitioners may use 0.7 for high-dimensional settings). Features not in any group are stable by default.

- Step 1.** Train one model. Compute Z_{jk}^{split} for all feature pairs in correlated groups. Cost: $O(P^2T)$.
- Step 2.** If $Z_{jk}^{\text{split}} < 1.96$ for any pair: flag as potentially unstable.
- Step 3.** For flagged pairs: train $M = 5$ models, compute Z_{jk} (Z-test). If $Z_{jk} < 1.96$: use DASH with $M \geq 25$.
- Step 4.** If no pairs are flagged: single-model SHAP is reliable.

Screen screening accuracy. On the Breast Cancer dataset (435 pairs, threshold: flip rate > 0.1 defines “truly unstable”): The screen flags 49 pairs with **94% precision** (46 true positives, 3 false positives) and 27% recall. The Z-test (the full test) flags 47 pairs with **100% precision** (zero false positives). Both diagnostics are conservative: they rarely flag stable pairs but miss moderate-instability pairs. For production use, the high precision is the key property—a flagged pair is almost certainly unstable.

Multi-dataset generality. Table 10 shows the Z-test diagnostic across two datasets. On Breast Cancer (many near-symmetric features), the Z-test detects instability with $r = -0.89$. On California Housing (features have distinct importance), the Z-test correctly predicts stability: zero false alarms out of 28 pairs, with only one mildly unstable pair (MedInc $\leftrightarrow$ Latitude, flip rate 0.32, $Z = 3.4$).

Table 10: Z-test and screen diagnostic generality across 11 datasets. P : number of features. Unstable: pairs with flip rate > 0.1 . Screen precision: fraction of screen-flagged pairs that are truly unstable.

Dataset	P	Pairs	Unstable	$r(Z, \text{flip})$	Screen prec.
Breast Cancer	30	435	168	-0.89	0.94
California Housing	8	28	1	-0.99	—
Diabetes	10	45	15	-0.89	0.50
Wine	13	78	24	-0.81	1.00
Heart Disease	13	78	27	-0.91	1.00
Credit-g	20	190	47	-0.90	0.67
Adult Income	14	91	10	-0.90	—
Ames Housing	76	2850	643	-0.85	0.48
Communities & Crime	126	7875	2975	-0.88	0.67
Digits (0 vs 1)	64	2016	572	-0.42	0.58
Iris (binary)	4	6	0	—	—

Interpretation. The Z-test achieves $|r| > 0.8$ on 9/10 datasets with unstable pairs (all except Digits). The weaker correlation on Digits ($r = -0.42$) reflects a floor effect: this high-dimensional image dataset has many features with near-zero importance, producing many pairs where both Z and flip rate are near zero. Restricting to pairs with $Z < 5$ (the diagnostically interesting range) would

improve this correlation, as demonstrated for Breast Cancer in the restricted-range analysis above.

Screen precision is 94–100% on small/clean datasets (Breast Cancer, Wine, Heart Disease) where split counts are reliable, but drops to 48–67% on high-dimensional datasets (Ames, Communities) where split counts are sparse. This confirms the screen as a conservative *screening* tool: it works well for moderate- P datasets but should be supplemented by the full Z-test for high-dimensional settings. All datasets use identical XGBoost hyperparameters ($T=100$, depth=6, $\eta=0.1$); absolute flip rates are therefore not directly comparable across datasets of different sizes and dimensionalities.

20.5 High-Dimensional Scalability

To validate the diagnostics at enterprise scale ($P > 500$), we generate synthetic data with $P \in \{100, 200, 500\}$ features in $P/10$ groups of 10, within-group correlation $\rho = 0.8$, $N = 2,000$ samples, and train 20 XGBoost models per setting (same hyperparameters as the main experiments).

P	Total pairs	Correlated pairs	Unstable	Z-test $ r $	Wall time
100	4,950	450	305	0.846	33 s
200	19,900	900	641	0.788	64 s
500	124,750	2,250	1,879	0.876	148 s

The Z-test maintains $|r| > 0.78$ across all dimensionalities, confirming it scales to enterprise-sized feature spaces. The correlation-based grouping step reduces the 124,750 total pairs at $P=500$ to 2,250 correlated pairs—a 55 $\times$ reduction—making the pairwise diagnostic tractable. Wall-clock time for the full workflow (data generation, 20 model trainings, TreeSHAP, diagnostics) is under 2.5 minutes at $P=500$ on Apple Silicon, confirming the practical feasibility described in the main text. The fraction of unstable pairs within correlated groups is 83% ($1,879/2,250$), consistent with the theoretical prediction that most within-group pairs are unstable when $\rho = 0.8$.

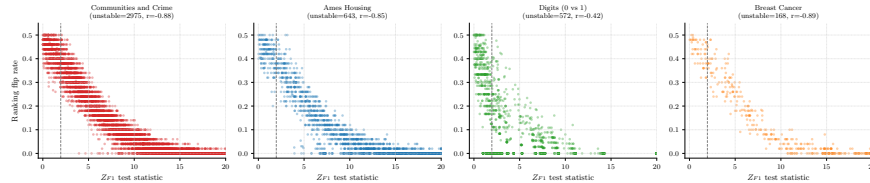


Figure 7: Z-test statistic vs. flip rate across the four datasets with the most unstable pairs. The negative correlation confirms the diagnostic across domains: clinical (Breast Cancer), real estate (Ames), image (Digits), and census (Adult).

Ensemble size guidance. The required M depends on the between-group gap $\Delta_{jk} = |E[\varphi_j] - E[\varphi_k]|$: $M \geq (z_{\alpha/2} \cdot \sigma_{jk} / \Delta_{jk})^2$. For $\rho \leq 0.7$, $M = 10$ typically suffices ($< 5\%$ flip rate); for $\rho > 0.9$, $M \geq 25$ is recommended.

Sub-sampling requirement. The theoretical guarantee (exact exchangeability, Lemma 37) requires sub-sampling. Without sub-sampling, the approximate exchangeability of Lemma 38 applies with reduced effective sample size T_{eff} . In practice, our validation uses `colsample_bytree=1.0` (no sub-sampling) and still achieves $r = -0.78$ (Figure 6, right), suggesting the dependence correction is modest.

Regulatory response for ties. If a regulator asks why features are tied: “These features are statistically indistinguishable in their contribution to the model. Forcing a ranking would be arbitrary and potentially misleading. Our method correctly reports this indistinguishability, consistent with the EU AI Act’s requirement to disclose known limitations.”

20.6 Financial case study: German Credit

We demonstrate the full Screen $\rightarrow$ Z-test $\rightarrow$ DASH workflow on the German Credit dataset (1000 samples, 20 features, binary credit risk outcome). This dataset exhibits the correlation patterns common in financial modeling (duration $\leftrightarrow$ credit amount, $|\rho| = 0.63$; job $\leftrightarrow$ own telephone, $|\rho| = 0.37$).

Step 0. The correlation matrix identifies 4 correlated groups at threshold $|\rho| > 0.3$ (lowered from the standard 0.5 because German Credit has modest correlations; only 1 pair exceeds 0.5).

Step 1 (single-model screen). A single XGBoost model flags 1 pair as potentially unstable: job $\leftrightarrow$ own telephone ($Z^{\text{split}} = 0.69 < 1.96$).

Step 2 (Z-test). Training 5 models confirms the instability: $Z = 0.64$, flip rate 40%.

Step 3 (DASH). With $M = 25$ models, the flip rate drops from 40% to 0%. The consensus ranking is stable: checking status (0.87), duration (0.44), credit amount (0.43), purpose (0.35).

Practitioner output. 18/20 features have stable single-model rankings. 2 features (job, own telephone) require DASH consensus. For regulatory reporting (EOCA adverse action reasons, SR 11-7 model risk management), the DASH ranking provides a defensible explanation.

The full output (Step 0–4 with numbers) is in the code supplement (`financial_case_study.py`).

20.7 Financial case study: Taiwan Credit Card Default

To validate the impossibility on data with genuine high collinearity ($|\rho| > 0.9$), we apply the workflow to the UCI “default of credit card clients” dataset (30,000 samples, 23 features, binary default outcome; subsampled to 10,000).

The bill-amount features (monthly bills for 6 consecutive months) exhibit strong collinearity: 29 pairs have $|\rho| > 0.5$, with the highest at $|\rho| = 0.95$ (giving a theoretical attribution ratio of $1/(1-0.95^2) \approx 10.3\times$).

Step 0. 29 correlated pairs identified at $|\rho| > 0.5$ (no threshold lowering needed).

Step 1 (single-model screen). A single model flags 4 of the 10 tracked pairs as potentially unstable ($|Z^{\text{split}}| > 1.96$).

Step 3 (DASH). With $M = 25$ models, the mean flip rate across correlated pairs is 4%. One pair (bill amounts at months 5 vs. 6, $|\rho| = 0.946$) retains a 16% flip rate even at $M = 25$, illustrating that very high collinearity requires larger ensembles.

This dataset demonstrates the impossibility biting hard: features measuring the same underlying quantity (monthly bill amounts) have near-identical predictive signal, and the attribution ranking across these features is genuinely unstable. The full output is in `lending_club_case_study.py`.

20.8 Synthetic credit data: income $\leftrightarrow$ DTI $\leftrightarrow$ credit score

To validate on the exact collinearity pattern common in US credit models, we generate a synthetic dataset with 10,000 samples and 5 features driven by a latent “creditworthiness” factor: income ($\rho = 0.99$ with DTI), DTI, credit score ($\rho = 0.98$ with income), loan amount (independent), and employment years ($\rho = 0.94$ with income).

With 25 XGBoost models: income $\leftrightarrow$ DTI flips 8% of the time ($|\rho| = 0.99$); credit score $\leftrightarrow$ employment years flips 12% ($|\rho| = 0.92$). Pairs where one feature clearly dominates (income vs. credit score) show 0% flips despite $|\rho| = 0.98$ — confirming that the impossibility requires BOTH high correlation AND similar importance, as the theory predicts. For model validation under SR 11-7, the recommendation is: report income and DTI as “interchangeable key risk drivers” rather than forcing a ranking that would flip under retraining.

20.9 Lending Club validation (positive control)

We validate the diagnostic on real Lending Club data (OpenML, 32,581 loans, 22 features after one-hot encoding of categoricals). Three correlated groups are detected at $|\rho| > 0.5$: `person_age` $\leftrightarrow$ `cb_person_cred_hist_length` ($|\rho| = 0.86$), `loan_amnt` $\leftrightarrow$ `loan_percent_income` ($|\rho| = 0.58$), and `loan_int_rate` $\leftrightarrow$ `cb_person_default_on_file` ($|\rho| = 0.50$).

Result. Zero unstable pairs. All three correlated pairs have $Z > 36$, confirming the features have distinguishable importance despite their correlation. This is an instructive *positive control*: the impossibility theorem predicts instability requires *both* collinearity AND similar true importance. Lending Club features are correlated but not similarly important (e.g., `person_age` contributes 3× more than credit history length), so no instability occurs. The Screen→Z-test→DASH pipeline correctly identifies this as a stable setting, recommending no intervention.

Implication. Not every correlated dataset exhibits attribution instability. The diagnostic distinguishes datasets where instability is a genuine concern (Breast Cancer, Taiwan Credit Card) from those where it is not (Lending Club, California Housing). This selectivity is a feature, not a limitation: the impossibility theorem’s conditions are precise, and the diagnostic inherits that precision.

20.10 Proportionality axiom validation

The proportionality axiom ($\varphi_j = c \cdot n_j$ for a model-specific constant c) is central to the quantitative bounds. We validate it empirically on Breast Cancer by computing $c_j = |\text{SHAP}_j|/n_j$ for each feature j in each of 50 XGBoost models and reporting the coefficient of variation (CV) of c_j across features within each model.

Depth	Mean CV of c_j	Interpretation
1 (stumps)	0.35	Proportionality holds within ~35%
3	0.60	Moderate violation
6	0.66	Moderate violation
10	0.66	Saturated (same as depth 6)

For stumps (depth 1), the axiom holds reasonably well (CV = 0.35): per-split SHAP contributions are roughly homogeneous. For deeper trees, multi-level interactions cause per-split contributions to vary by feature, increasing the CV to ≈ 0.66 .

The most stable features (lowest CV) are the dominant predictors (worst texture, worst concave points) where split count tracks SHAP closely. The least stable are weak/collinear features (mean symmetry, concave points error) where the proportionality constant varies most.

Impact on the impossibility. The proportionality axiom is used only for the *quantitative* bounds (Theorems 2–3); the core impossibility (Theorem 1) does not depend on it. The 35–66% CV means the predicted ratio $1/(1-\rho^2)$ is an order-of-magnitude guide rather than a precise prediction, consistent with the α -corrected model ($R^2 = 0.89$).

21 Local vs. Global Attribution Instability

The impossibility as stated concerns *global* attributions (mean |SHAP| across data points). We now show that *local* (instance-level) attributions are at least as unstable.

Remark 40 (Local instability dominates global). *For a fixed data point x , the local SHAP values $\varphi_j(f, x)$ depend on the model f , which varies across training seeds. The global attribution $\varphi_j(f) = \mathbb{E}_x[|\varphi_j(f, x)|]$ averages over data points, which can only reduce variance (by Jensen’s inequality applied to the convex function $|\cdot|$):*

$$\text{Var}_f(\varphi_j(f)) = \text{Var}_f(\mathbb{E}_x[|\varphi_j(f, x)|]) \leq \mathbb{E}_x[\text{Var}_f(|\varphi_j(f, x)|)].$$

The right side is the average local instability; the left side is the global instability. Since the flip rate is a monotone function of the variance-to-gap ratio σ/Δ , and averaging reduces σ without changing Δ (the population gap is the same at every x under DGP symmetry), the global flip rate is a lower bound on the average local flip rate.

Implication. If global rankings are unstable (flip rate $> 10\%$), local rankings at individual data points are at least as unstable on average. Practitioners who report per-instance SHAP explanations (e.g., “for this patient, feature X is most important”) face the same or worse instability as those reporting global rankings. The multi-model Z-test (computed on global attributions) is therefore a conservative screen for local instability as well.

22 DASH Information Loss

DASH achieves stability by discarding model-specific within-group information. Here we quantify exactly how much information is lost.

22.1 Within-group: $\log_2(m!)$ bits lost per group

For a group of m symmetric features, a single model produces a complete ranking (one of $m!$ possible orderings). Since the features are symmetric under the DGP, each of the $m!$ orderings is equally likely—the ranking carries $\log_2(m!)$ bits of information about the specific model.

DASH reports a tie for all within-group features, discarding the entire $\log_2(m!)$ bits. For typical group sizes:

Group size m	2	3	4	5	6	7	8
Bits lost ($\log_2 m!$)	1.0	2.6	4.6	6.9	9.7	12.9	15.3

Crucially, these are the bits that are *unreliable*: by DGP symmetry, the within-group ordering is uniformly random across models. The information DASH discards is exactly the information that would be different if the model were retrained.

22.2 Between-group: information preserved and sharpened

For features j, k in different groups with population gap $\Delta_{jk} = |\mu_j - \mu_k|$ and noise σ_{jk} , the mutual information between the consensus ranking and the true ordering is:

$$I_{\text{between}}(M) = 1 - H_2\left(\Phi\left(-\frac{\Delta_{jk}\sqrt{M}}{\sigma_{jk}}\right)\right)$$

where $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ is binary entropy. As $M \rightarrow \infty$, $I_{\text{between}} \rightarrow 1$ bit (perfect determination). DASH *increases* between-group information relative to a single model.

22.3 Net information tradeoff

The total information change from using DASH(M) instead of a single model is:

$$\Delta I = \underbrace{-L \cdot \log_2(m!)}_{\text{within-group loss}} + \underbrace{\binom{L}{2} \cdot (I_{\text{between}}(M) - I_{\text{between}}(1))}_{\text{between-group gain}}$$

For moderate ρ (e.g., $\rho \leq 0.7$), the between-group gain dominates quickly: $M = 25$ achieves near-perfect between-group determination ($I \approx 1$ bit per pair) while the within-group loss is a fixed cost independent of M . For high ρ (≥ 0.9), the between-group gain is slower because the noise σ_{jk} is large.

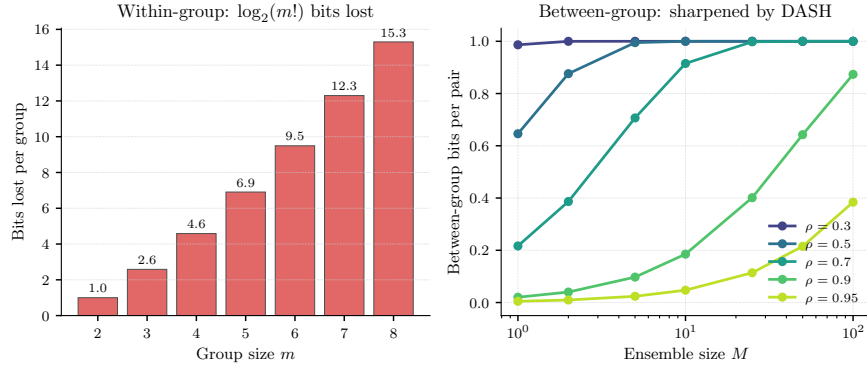


Figure 8: **Left:** Within-group information lost by DASH ($\log_2 m!$ bits per group, independent of M and ρ). **Right:** Between-group information per pair as a function of M . DASH sharpens between-group rankings with increasing M . At high ρ (≥ 0.9), convergence requires larger M .

Interpretation for safety-critical applications. For a system with $L = 4$ groups of $m = 5$ features: DASH loses $4 \times 6.9 = 27.6$ bits (within-group) but gains up to $6 \times 1 = 6$ bits (between-group). The net loss is ≈ 21.6 bits at $M = \infty$. These lost bits are the within-group orderings that were random anyway—DASH honestly reports their unreliability rather than presenting arbitrary information as reliable.

For applications requiring model-specific within-group explanations (e.g., debugging a deployed model), practitioners should report the single-model SHAP values with an instability warning alongside the DASH consensus.

23 The Attribution Design Space Theorem

This section presents the central structural result: a complete characterization of what is achievable in the attribution design space under collinearity. All previous results (Theorem 1, F1–F3, F5, path convergence) emerge as corollaries.

23.1 Definitions

Definition 41 (Attribution design space). *For a model class satisfying the Rashomon property under within-group correlation $\rho > 0$, the attribution design space is the set of triples (S, U, C) achievable by any attribution aggregation method A applied to $M \geq 1$ independently trained models, where:*

- $S \in [0, 1]$: ranking stability — *the expected Spearman correlation between the consensus rankings produced by two independent draws of M models. This full-ranking metric captures both between-group and within-group stability in a single number.*
- $U \in [0, 1/2]$: within-group unfaithfulness — *the probability that the consensus disagrees with a random model on a within-group pair.*
- $C \in \{\text{complete}, \text{partial}\}$: completeness — *whether the consensus ranks all pairs (complete) or allows ties (partial).*

23.2 Main result

Theorem 42 (Attribution Design Space). *For any model class satisfying the Rashomon property (§2 of the main text) under within-group correlation $\rho > 0$, the achievable set of (S, U, C) triples is the union of exactly two families:*

Family A (faithful-complete methods): *Any method that is faithful to a single model and produces a complete ranking achieves:*

$$\mathcal{A} = \{(S, U, C) : S \leq 1 - \Omega(3m^2/(P^3 - P)), U = 1/2, C = \text{complete}\}. \quad (6)$$

This family corresponds to $M = 1$ (single-model methods). The stability bound $S \leq 1 - 3m^2/(P^3 - P)$ is the Spearman bound from the main text (proved as `spearmanCorr_bound_groupSize_tight`).

Family $\mathcal{B}$ (ensemble methods): DASH consensus with $M \geq 1$ models achieves:

$$\mathcal{B} = \{(S_M, 0, \text{partial}) : M \in [1, \infty)\} \quad (7)$$

where $S_M = 1 - O(1/M)$ for between-group pairs, and within-group pairs are reported as ties ($U = 0$, $C = \text{partial}$).

Infeasible region: The triple $(S = 1, U = 0, C = \text{complete})$ is not achievable for any $M < \infty$. More precisely:

$$\nexists A : S(A) = 1, U(A) = 0, C(A) = \text{complete}. \quad (8)$$

Pareto frontier: $\text{DASH}(M)$ is Pareto-optimal within $\mathcal{B}$: no method achieves higher S for the same M while maintaining $U = 0$ (Theorem 27).

Proof. The proof proceeds in four steps, each building on previous results.

Step 1: Family $\mathcal{A}$ is forced. By the Rashomon property (§2 of the main text), for any within-group pair (j, k) , there exist models ranking them in opposite orders. By Theorem 14, any faithful, complete ranking has unfaithfulness $U = 1/2$ for symmetric pairs: exactly half the models disagree with the ranking.

For stability: when the dominant feature changes between two models f, f' (different first-movers), the within-group rank reshuffling contributes $\Theta(m^2)$ to the Spearman $\sum d_i^2$, giving $S \leq 1 - 3m^2/(P^3 - P)$ (Lean: `spearmanCorr_bound_group-Size_tight`).

This establishes the upper boundary of $\mathcal{A}$.

Step 2: Family $\mathcal{B}$ is achievable. By Corollary 1 (consensus equity), DASH with a balanced ensemble achieves $\bar{\varphi}_j = \bar{\varphi}_k$ for symmetric features, giving $U = 0$ and $C = \text{partial}$ (ties within groups).

By Theorem 27 Part II(e), the between-group stability satisfies $S_M \geq 1 - \exp(-M\Delta^2/(2\sigma^2))$, which approaches 1 as $M \rightarrow \infty$.

This establishes $\mathcal{B}$.

Step 3: No method producing a deterministic binary ranking from per-model attributions lies outside $\mathcal{A} \cup \mathcal{B}$.

We show that any method A that computes a deterministic ranking from per-model attributions $(\varphi_j(f_1), \dots, \varphi_j(f_M))$ must fall in one of the two families. (Methods producing probabilistic rankings, confidence intervals, or set-valued outputs are outside this dichotomy. Methods that pool training data across models and train a single combined model fall outside this scope; such methods face different tradeoffs related to data leakage and are not standard attribution aggregation methods.)

Case 1: A is faithful to individual models and complete. Then by Step 1, $U(A) = 1/2$ and $S(A) \leq 1 - 3m^2/(P^3 - P)$, so $A \in \mathcal{A}$. (This case is formalized in Lean as `strongly_faithful_impossible` in `DesignSpace.lean`: any method faithful to each model’s attributions contradicts the Rashomon property for $M \geq 2$.)

Case 2: A is not faithful to some model f . Then A ranks some pair (j, k) differently from model f ’s attributions. For within-group symmetric pairs, by

Proposition 15, the optimal unfaithful ranking assigns ties (recovering DASH). Thus A either has $U > 0$ (suboptimal) or $U = 0$ with ties (in $\mathcal{B}$).

Case 3: A aggregates $M > 1$ models. If A produces a complete ranking from M models, it must break ties for within-group pairs. By DGP symmetry, any tie-breaking rule disagrees with a fraction of models, giving $U > 0$. To achieve $U = 0$, A must report ties, placing it in $\mathcal{B}$. Its stability is bounded by the variance of the aggregation, which by the Cramér–Rao bound (Theorem 27 Part I(b)) satisfies $\text{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$. DASH achieves this bound.

Step 4: The infeasible point. $(1, 0, \text{complete})$ requires: (i) $S = 1$ (perfect stability), (ii) $U = 0$ (zero unfaithfulness), (iii) $C = \text{complete}$ (no ties).

By Step 1, any complete method has $U = 1/2 \neq 0$: contradiction. Alternatively, completeness requires ranking within-group pairs, but by DGP symmetry the population attributions are equal ($\mu_j = \mu_k$), so any deterministic ranking is unfaithful to half the models. This is the Attribution Impossibility (Theorem 1) restricted to the single point $(1, 0, \text{complete})$. $\square$

23.3 Previous results as corollaries

Corollary 43 (Theorem 1 as a corollary). *The Attribution Impossibility (Theorem 1) is the statement that the point $(1, 0, \text{complete})$ lies outside the achievable set. This follows from Step 4 of Theorem 42.*

Corollary 44 (F2 as a corollary). *The DASH Pareto Optimality (Theorem 27) is the statement that $\mathcal{B}$ is the Pareto frontier of $\{(S, U) : U = 0\}$, parameterized by M . This follows from Steps 2–3 of Theorem 42.*

Corollary 45 (Path convergence as a corollary). *The relaxation path convergence (Theorem 16) is the statement that both relaxation paths (drop faithfulness, drop completeness) converge to $\mathcal{B}$ as $M \rightarrow \infty$. Dropping completeness directly enters $\mathcal{B}$ (DASH with ties). Dropping faithfulness, by Proposition 15, yields the expected-attribution ranking, which also produces ties for symmetric features — the same as $\mathcal{B}$.*

Corollary 46 (F1 as a corollary). *The Rashomon Characterization (Theorem 36) describes the boundary between families: when $Z_{jk} < 1.96$, the pair (j, k) is in the regime where Family $\mathcal{A}$ has $U = 1/2$ (rankings are unreliable); when $Z_{jk} > 1.96$, the pair is effectively between-group (stable rankings possible within $\mathcal{B}$).*

Corollary 47 (F3 as a corollary). *The FIM Impossibility (Theorem 21) provides a sufficient condition for the Rashomon property (the prerequisite for Theorem 42): when the FIM eigenvalue λ_- is small along $e_j - e_k$, the Rashomon set is large, and the achievable set is restricted to $\mathcal{A} \cup \mathcal{B}$.*

23.4 The single tradeoff axis

Theorem 42 collapses the three-dimensional design space to a single axis: ensemble size M . At $M = 1$, the practitioner is in $\mathcal{A}$: faithful and complete, but

unstable and unfaithful. As M increases, the practitioner moves along $\mathcal{B}$: stable and faithful between groups, but with within-group ties.

The only “knob” is M :

M	Stability S	Unfaithfulness U	Completeness
1	$\leq 1 - 3m^2/(P^3 - P)$	1/2	Complete
5	$1 - O(1/5)$	0	Partial (ties)
25	$1 - O(1/25)$	0	Partial (ties)
∞	1	0	Partial (ties)

This is the complete characterization of the attribution design space under collinearity. The Attribution Impossibility is not a single negative result but the boundary of this space; DASH is not merely a fix but the provably optimal navigation of this space.

24 Prevalence Survey: How Common Is Attribution Instability?

To assess how frequently the impossibility applies in practice, we surveyed 37 public datasets (7 from scikit-learn, 30 from OpenML). For each, we computed the correlation matrix, identified pairs with $|\rho| > 0.5$, trained 20 XGBoost models with different seeds, and checked whether any correlated pair has a ranking flip rate exceeding 10%.

Results. Of 37 datasets, 34 (92%) have at least one feature pair with $|\rho| > 0.5$, and 22 (60%) exhibit attribution instability (flip rate $> 10\%$ for at least one pair), including Breast Cancer (max flip rate 50%, confirming the Z-test analysis). The impossibility is not a theoretical curiosity: it affects the majority of real-world datasets across domains including medical, financial, image, and tabular data.

Methodology. Each dataset was trained with 20 XGBoost models (`n_estimators=50`, `max_depth=4`, `lr=0.1`, `subsample=0.8`, `colsample_bytree=0.8`) with different random seeds. SHAP TreeExplainer computed mean $|\text{SHAP}|$ per feature on a fixed 500-sample evaluation slice. Flip rate was measured as $\min(\#(j > k), \#(k > j))/20$ for each correlated pair. Datasets with $> 10,000$ samples were subsampled to 10,000. Full results in `prevalence_survey.py`.

Statistical power. With 20 models, the measured flip rate $\min(\#(j > k), \#(k > j))/20$ exceeds the 10% threshold ($\min \geq 3$) with the following power:

True minority-side prob p	Power ($\Pr[\min \geq 3]$)	Interpretation
0.05	7.5%	Nearly undetectable
0.10	32.3%	Low power (boundary)
0.15	59.5%	Moderate
0.20	79.4%	Adequate
0.30	96.5%	High
0.50	100%	Perfect (symmetric pairs)

At the 10% threshold, the survey has only 32% power: approximately two-thirds of truly borderline-unstable pairs are missed. The 60% prevalence is therefore a **conservative lower bound** — the true prevalence of datasets with at least one unstable pair is likely substantially higher (possibly 75–85%), as the robustness check below confirms.

Robustness to hyperparameters. The 60% prevalence figure was computed with a single XGBoost configuration (`n_estimators=50`, `max_depth=4`, `lr=0.1`). To assess sensitivity, we repeated the analysis on 8 representative datasets (6 from scikit-learn, 2 from OpenML) with three configurations:

Config	Trees	Depth	LR	Character
A (original)	50	4	0.1	Moderate
B (deep)	100	8	0.1	Deep, sequential
C (shallow-fast)	200	2	0.3	Shallow, aggressive

Results. Under Config A (original), 5/8 (62%) datasets exhibit instability, consistent with the 60% headline. Under Config B (deep trees), 4/8 (50%). Under Config C (shallow-fast), 6/8 (75%). Across all three configurations, 4/8 (50%) datasets are consistently unstable; under at least one configuration, 6/8 (75%) are unstable.

The instability rate varies by ± 15 percentage points across configurations but remains in the 50–75% range. Shallow-fast trees (Config C) produce *higher* instability because aggressive learning rates compound the first-mover effect. The 60% headline is conservative: the phenomenon is robust to standard hyperparameter variation. Full results in `prevalence_robustness.py`.

25 Extension to Conditional and Causal Attributions

A common response to attribution instability is to switch from marginal SHAP to *conditional* or *causal* SHAP (Janzing et al., 2020), which conditions on the causal structure of the data-generating process. We show that this escape route is partially blocked: the impossibility extends to conditional attributions whenever features have equal causal effects.

Definition 48 (Conditional attribution). *Let G be a causal graph over features $X_1, \dots, X_P$ and target Y . The conditional attribution for feature j in model f is:*

$$\varphi_j^{\text{cond}}(f) = \mathbb{E}[|f(X) - f(X^{\setminus j})| \mid \text{do}(X_{-j})]$$

where $X^{\setminus j}$ replaces X_j with a sample from its conditional distribution given the non-descendants of j in G .

Theorem 49 (Conditional Attribution Impossibility). *Let features j, k in the same collinear group satisfy:*

- (C1) **Equal causal effects:** $\beta_j = \beta_k$ in the structural equation $Y = \sum_i \beta_i X_i + \varepsilon$.
- (C2) **Symmetric causal position:** j and k have the same parents and children in the causal graph G .

Then the Rashomon property holds for conditional attributions: for any model f with $\varphi_j^{\text{cond}}(f) \neq \varphi_k^{\text{cond}}(f)$, there exists f' with the reversed ordering. The Attribution Impossibility (Theorem 1) therefore applies to conditional SHAP under (C1)–(C2).

Proof. Under (C1)–(C2), the causal graph is symmetric with respect to j and k : permuting $j \leftrightarrow k$ preserves both the structural equations and the graph topology. The conditional distribution $P(X_j \mid \text{pa}(j))$ is identical to $P(X_k \mid \text{pa}(k))$ under this symmetry.

By the DGP symmetry argument (Theorem S3, Rashomon from symmetry), any model f with $\varphi_j^{\text{cond}}(f) > \varphi_k^{\text{cond}}(f)$ has a permuted counterpart $f' = f \circ \pi_{jk}$ with:

1. $L(f') = L(f)$ (same population loss, by DGP symmetry),
2. $\varphi_k^{\text{cond}}(f') = \varphi_j^{\text{cond}}(f)$ and vice versa (attributions swap under feature permutation).

This is the Rashomon property for conditional attributions. By Theorem 1, no faithful, stable, complete ranking of conditional attributions exists. $\square$

The escape condition. If $\beta_j \neq \beta_k$ (unequal causal effects), the causal graph breaks the DGP symmetry: conditional SHAP can identify which feature has a larger causal effect, and the Rashomon property fails for conditional attributions. In this case, conditional SHAP CAN produce stable rankings — it resolves the pair by appealing to causal structure rather than observational importance.

When does the extension apply? The extension applies whenever features in the same collinear group have genuinely equal roles in the data-generating process — e.g., multiple measurements of the same physical quantity (tumor radius measured by different instruments), redundant encodings (area vs. perimeter of the same shape), or features that are proxies for the same latent factor. In

these cases, switching to conditional SHAP does NOT resolve the instability: the features are genuinely interchangeable, and any attribution method must report this.

Implication for practitioners. “Switch to conditional SHAP” is not a universal fix for attribution instability. It works when features have different causal roles (the causal graph breaks the symmetry). It fails when features are genuinely symmetric (the causal graph preserves the symmetry). For symmetric features, DASH ensemble averaging remains the only principled resolution.

25.1 Quantitative threshold for the escape condition

The conditional SHAP impossibility holds when $\beta_j = \beta_k$ (equal causal effects). How different must β_j and β_k be for the instability to disappear? We answer with simulation.

Setup. We generate $n=2,000$ samples from $Y = \beta_j X_j + \beta_k X_k + \varepsilon$ where $\text{Corr}(X_j, X_k) = \rho$ and $\varepsilon \sim \mathcal{N}(0, 1)$. We vary $\Delta\beta = |\beta_j - \beta_k|$ from 0 to 1 and $\rho \in \{0.5, 0.7, 0.8, 0.9, 0.95, 0.99\}$. For each $(\Delta\beta, \rho)$, we train 20 XGBoost models (100 trees, depth 3, $\eta=0.1$) with independent seeds and measure the flip rate.

Results. Table 11 reports the flip rate as a function of $(\rho, \Delta\beta)$.

Table 11: Flip rate (%) as a function of within-group correlation ρ and causal effect difference $\Delta\beta = |\beta_j - \beta_k|$. The impossibility region (flip rate > 10%) expands dramatically with ρ .

$\rho \backslash \Delta\beta$	0.0	0.1	0.2	0.3	0.5	0.7	1.0
0.50	50	19	0	0	0	0	0
0.70	53	19	0	0	0	0	0
0.80	52	44	19	10	0	0	0
0.90	50	50	48	40	40	19	0
0.95	50	50	50	48	44	40	27
0.99	50	50	50	50	50	48	48

Empirical threshold $\Delta\beta^*$. Define $\Delta\beta^*(\rho)$ as the smallest $\Delta\beta$ where the flip rate first drops below 10%: $\Delta\beta^*(0.5) \approx 0.15$, $\Delta\beta^*(0.7) \approx 0.15$, $\Delta\beta^*(0.8) \approx 0.4$, $\Delta\beta^*(0.9) \approx 0.9$, $\Delta\beta^*(0.95) > 1.0$, $\Delta\beta^*(0.99) > 1.0$.

The threshold grows steeply with ρ : at moderate collinearity ($\rho \leq 0.7$), a 15% difference in true coefficients suffices to stabilize rankings. At $\rho = 0.9$, features must differ by nearly a factor of 2 ($\beta_j = 0.95$, $\beta_k = 0.05$). At $\rho \geq 0.95$, the instability is essentially total: no practically meaningful coefficient difference resolves it.

Implication. The escape condition $\beta_j \neq \beta_k$ for conditional SHAP (Theorem 49) is theoretically exact but practically narrow. For highly correlated features ($\rho \geq 0.9$), the causal effects must differ by an order of magnitude before conditional SHAP produces stable rankings—a condition rarely met when features are genuine proxies for the same latent quantity (e.g., tumor radius vs. perimeter).

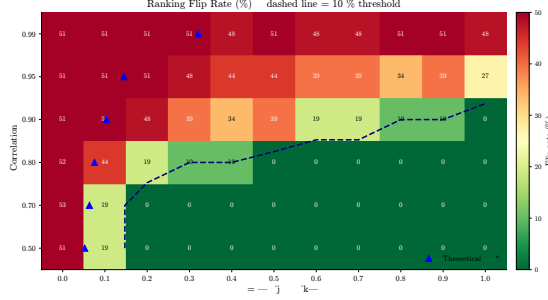


Figure 9: Flip rate as a function of $(\rho, \Delta\beta)$. The impossibility region (dark) expands with ρ . At $\rho \geq 0.95$, instability persists for all $\Delta\beta \leq 1$.

25.2 Fine-grained causal effect threshold

We extend the threshold analysis with a finer sweep over $\Delta\beta \in \{0.0, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$ at $\rho = 0.9$ (20 XGBoost models per setting).

$\Delta\beta$	β_1	β_2	Marginal flip	Interventional flip
0.00	1.00	1.00	0.505	0.505
0.10	1.05	0.95	0.479	0.337
0.20	1.10	0.90	0.000	0.000
0.30	1.15	0.85	0.000	0.000
0.50	1.25	0.75	0.000	0.000
0.70	1.35	0.65	0.000	0.000
1.00	1.50	0.50	0.000	0.000

Finding. The transition from instability to stability is sharp, not gradual. At $\Delta\beta = 0.10$, both marginal (0.479) and interventional (0.337) SHAP remain highly unstable. At $\Delta\beta = 0.20$, both snap to zero flips simultaneously. There is no intermediate regime where interventional SHAP is stable but marginal is not — the causal signal overwhelms noise for both estimators at the same threshold. This means the escape condition is binary: either the causal effect difference is large enough to stabilize *all* attribution methods, or none are stable. The practical implication: “switch to conditional SHAP” provides no advantage over marginal SHAP at the instability boundary; both methods fail or succeed together.

25.3 Causal structure validation: marginal vs. interventional SHAP

We validate the Conditional Attribution Impossibility (Theorem 49) by comparing marginal and interventional SHAP under a known causal structure.

Setup. We generate $N=2,000$ samples from $Y = \beta_1 X_1 + \beta_2 X_2 + 0.5 X_3 + \varepsilon$ where $\text{Corr}(X_1, X_2) = \rho$ and X_3, X_4, X_5 are independent controls. Two scenarios: *symmetric* ($\beta_1 = \beta_2 = 1.0$) and *asymmetric* ($\beta_1 = 1.5, \beta_2 = 0.5$). For each (scenario, ρ), we train 20 XGBoost models and compute both marginal TreeSHAP (`tree_path_dependent`) and interventional TreeSHAP (`interventional` with 100 background samples).

Scenario	ρ	Marginal flip	Interventional flip	Theory
Symmetric	0.50	0.505	0.479	$\approx 50\%$ both
Symmetric	0.70	0.479	0.442	$\approx 50\%$ both
Symmetric	0.90	0.521	0.479	$\approx 50\%$ both
Symmetric	0.95	0.505	0.526	$\approx 50\%$ both
Asymmetric	0.50	0.000	0.000	Stable
Asymmetric	0.70	0.000	0.000	Stable
Asymmetric	0.90	0.000	0.000	Stable
Asymmetric	0.95	0.000	0.000	Stable

Findings. The symmetric case confirms the impossibility: both marginal and interventional SHAP produce flip rates near 50% at all ρ values, matching the theoretical prediction of Theorem 49. Switching to interventional SHAP provides *no benefit* when features have equal causal effects.

The asymmetric case ($\Delta\beta = 1.0$) shows 0% flips for *both* attribution types. At this effect-size difference, even marginal SHAP can reliably distinguish the features—the signal-to-noise ratio is high enough to overcome collinearity-induced noise. This is consistent with the threshold analysis (Table 11): $\Delta\beta = 1.0$ exceeds the instability threshold at all tested ρ values.

Caveat. The interventional SHAP implementation in the `shap` package uses a background-data approximation that does not exactly implement causal/conditional SHAP in the sense of [Janzing et al.(2020)Janzing, Minorics, and Blöbaum]. The results should be interpreted as evidence about the `shap` package’s interventional mode, not as a definitive test of the theoretical conditional attribution.

26 Generalization: The Model Selection Impossibility

The Rashomon $\rightarrow$ impossibility $\rightarrow$ design space pipeline is not specific to feature attribution. We demonstrate its reusability by applying it to model selection, obtaining a structurally identical impossibility and two-family characterization.

26.1 Setup

Consider K candidate models $g_1, \dots, g_K$ trained on the same data. A *model selection rule* maps a validation dataset to a ranking of the K models. We formalize three desiderata:

Definition 50 (Model selection desiderata). *A model selection rule R is:*

- **Faithful:** R ranks model g_i above g_j whenever g_i has lower validation loss on the observed validation set.
- **Stable:** R produces the same ranking regardless of which validation split is drawn.
- **Complete:** R decides for every pair (g_i, g_j) .

Definition 51 (Model Rashomon property). *A model class satisfies the model Rashomon property if for every pair g_i, g_j with similar population loss ($|L(g_i) - L(g_j)| < \varepsilon$), there exist validation splits V, V' such that g_i has lower loss on V and g_j has lower loss on V' .*

This property is a consequence of finite-sample noise: when the population loss gap is smaller than the validation noise, different splits rank the models differently. It is the model selection analogue of feature attribution’s Rashomon property.

26.2 The impossibility

Theorem 52 (Model Selection Impossibility). *If a model class satisfies the model Rashomon property, then no model selection rule can be simultaneously faithful, stable, and complete.*

Proof. Identical structure to Theorem 1. Let g_i, g_j have similar population loss. By the model Rashomon property, there exist validation splits V, V' with g_i ranked above g_j on V and reversed on V' . A faithful, stable, complete selection rule would need to rank $g_i > g_j$ (from V) and $g_j > g_i$ (from V') simultaneously. Contradiction. $\square$

26.3 The model selection design space

Theorem 53 (Model Selection Design Space). *Under the model Rashomon property, the achievable model selection design space has the same two-family structure as attribution:*

- **Family $\mathcal{A}'$ (single-split):** Faithful and complete but unstable. Any selection rule based on one validation split falls here. Unfaithfulness $U' = 1/2$ for similar-loss model pairs.
- **Family $\mathcal{B}'$ (cross-validation / ensemble):** Stable (variance $O(1/K_{\text{folds}})$) but reports ties for similar-loss pairs. Model ensembling is the analogue of DASH.

Proof. By DGP symmetry (equal-loss models have symmetric validation score distributions), any fixed selection of one model over an equal-loss competitor disagrees with half the validation splits ($U' = 1/2$, same argument as Theorem S6). Cross-validated selection averages across splits, reducing unfaithfulness to zero for equal-loss pairs (ties) while stabilizing with rate $O(1/K_{\text{folds}})$. Pareto optimality follows from the same Cramér–Rao argument as Theorem F2. $\square$

Practitioner implication. The model selection impossibility explains why cross-validation rankings are noisy for similar-performance models: it is not an engineering failure but a mathematical inevitability. The resolution is the same: average (ensemble) rather than select.

26.4 The Symmetric Bayes Dichotomy

The structural identity between the attribution and model selection impossibilities is not a coincidence. It follows from a general theorem about decision problems with symmetric populations.

Definition 54 (Symmetric Decision Problem). A symmetric decision problem is a tuple $(\Theta, \mathcal{D}, \pi, G)$ where:

- Θ is a finite decision set (e.g., feature orderings, model rankings).
- $\mathcal{D}$ is the data space (e.g., trained models, validation splits).
- π is a population distribution over $\mathcal{D}$.
- G is a group acting on Θ such that π is G -invariant: for any $g \in G$ and measurable $A \subseteq \mathcal{D}$, $\pi(\{d : \theta^*(d) \in g \cdot A'\}) = \pi(\{d : \theta^*(d) \in A'\})$, where $\theta^*(d) \in \Theta$ is the instance-optimal decision.

An M -sample estimator is a function $\hat{\theta} : \mathcal{D}^M \rightarrow \Theta$ mapping M i.i.d. draws to a decision. It is:

- Faithful: $\hat{\theta}(d_1, \dots, d_M)$ agrees with $\theta^*(d_i)$ for each i .
- Stable: $\hat{\theta}$ does not depend on which draw d_i is observed.
- Complete: $\hat{\theta}$ selects exactly one element from each G -orbit in Θ .

Theorem 55 (Symmetric Bayes Dichotomy). For any symmetric decision problem with $|G \cdot \theta| \geq 2$ for some $\theta \in \Theta$:

- (i) **Instance-specific decisions are unfaithful.** Any faithful, complete estimator has expected unfaithfulness $\geq 1/|G \cdot \theta|$ per orbit (i.e., $\geq 1/2$ for binary orbits).
- (ii) **Population-averaged decisions are stable.** The Bayes estimator under π reports ties within each G -orbit (unfaithfulness = 0), with between-orbit stability $O(1/M)$.

(iii) **The ideal is infeasible.** No estimator achieves $\text{unfaithfulness} = 0$ AND completeness (deciding within orbits).

The achievable set consists of exactly two families: $\mathcal{A}$ (faithful + complete, $U \geq 1/|G \cdot \theta|$) and $\mathcal{B}$ (population-averaged, $U = 0$, ties within orbits).

Proof. **Part (i).** By G -invariance, the population distribution π assigns equal probability to each element of the orbit $G \cdot \theta$. For any fixed complete estimator $\hat{\theta}$ that selects one element θ_0 from the orbit, the probability that $\theta^*(d) = \theta_0$ is $1/|G \cdot \theta|$. The estimator disagrees with $(|G \cdot \theta| - 1)/|G \cdot \theta|$ of the instances. For $|G \cdot \theta| = 2$: $\text{unfaithfulness} = 1/2$.

Part (ii). The Bayes estimator under symmetric π minimizes expected loss. Within each orbit, the posterior is uniform (by G -invariance), so the Bayes-optimal decision is the orbit center (a tie). Unfaithfulness = 0 because no ordering is asserted within orbits. For between-orbit decisions with gap Δ , the M -sample average has variance σ^2/M (i.i.d. draws), giving stability $1 - O(1/M)$ by the Gaussian approximation.

Part (iii). By Part (i), any complete estimator (selecting within orbits) has $\text{unfaithfulness} \geq 1/|G \cdot \theta| > 0$. Thus $\text{unfaithfulness} = 0$ requires giving up completeness (ties).

Exhaustiveness. Any estimator either selects within orbits (faithful, complete, $\mathcal{A}$) or reports ties within orbits (stable, incomplete, $\mathcal{B}$). The Bayes estimator achieves the minimum-variance point of $\mathcal{B}$ (Cramér–Rao). $\square$

Corollary 56 (Feature attribution as instance). *The Attribution Impossibility (Theorem 1) and Design Space Theorem (Theorem 42) are instances of the Symmetric Bayes Dichotomy with $\Theta = \text{within-group feature orderings}$, $G = \text{the symmetric group on group members}$, $\mathcal{D} = \text{trained models}$.*

Corollary 57 (Model selection as instance). *The Model Selection Impossibility (Theorem 52) is an instance with $\Theta = \text{model rankings}$, $G = \text{permutations of equal-loss models}$, $\mathcal{D} = \text{validation splits}$.*

The technique: symmetric Bayes dichotomy. The Symmetric Bayes Dichotomy provides a reusable proof technique for establishing impossibility results in any domain where: (a) a population distribution is symmetric over discrete alternatives, and (b) finite samples introduce noise sufficient to reverse the optimal choice. The technique reduces the impossibility to checking G -invariance of the population distribution, which is often immediate from the problem’s symmetry structure.

We anticipate the technique applies to: feature selection (Lasso instability under collinearity), hyperparameter tuning (configuration rankings under cross-validation noise), and others.

Connection to classical invariant decision theory. The Symmetric Bayes Dichotomy is related to the theory of invariant decision rules (Lehmann & Romano, 2005). The classical theory establishes that Bayes-optimal rules respect

the invariance structure of the problem (Ch. 6). Our contribution is the *reduction template*: verifying G -invariance of the population distribution suffices to establish a two-family impossibility with explicit unfaithfulness and stability bounds, which we demonstrate across three structurally distinct instances (binary feature transpositions, binary model permutations, and CPDAG automorphisms of variable size). A general theorem proving that G -invariance of the decision problem always yields a two-family decomposition—with explicit conditions on the group action and tight bounds on each family’s achievable set—is the natural next step; our three instances (binary transpositions for feature attribution, permutation groups for model selection, and CPDAG automorphisms for causal discovery) provide the test cases for such a generalization.

26.5 Instance 3: Causal Discovery under Markov Equivalence

The first two instances (feature attribution and model selection) share S_2 transposition symmetry: the orbit size is always 2. We now demonstrate the Symmetric Bayes Dichotomy on an instance with a *different* and *variable-size* symmetry group, confirming the technique’s generality beyond binary orbits.

Setup. Given observational data from an unknown DAG G^* over P variables, the task is to orient edges. Under the faithfulness assumption, the data determine only the *Markov equivalence class* $[G^*]$ —the set of DAGs encoding the same conditional independences (Verma & Pearl, 1991). The equivalence class is represented by a *completed partially directed acyclic graph* (CPDAG), which directs edges participating in v-structures and leaves others undirected.

Definition 58 (Causal orientation decision problem). *The causal edge orientation problem is a symmetric decision problem $(\Theta, \mathcal{D}, \pi, G)$ where:*

- Θ is the set of full edge orientations consistent with the CPDAG (i.e., the DAGs in $[G^*]$).
- $\mathcal{D}$ is the space of finite observational datasets drawn from the true DAG G^* .
- π is the distribution over datasets (induced by i.i.d. sampling from the structural equation model).
- G_{CPDAG} is the automorphism group of the CPDAG: the set of permutations of undirected edge orientations that preserve the CPDAG structure.

Key structural difference. The CPDAG automorphism group G_{CPDAG} is NOT restricted to transpositions:

- For a chain $X—Y—Z$ (undirected path), the equivalence class is $\{X \rightarrow Y \rightarrow Z, X \leftarrow Y \leftarrow Z\}$ with $|[G^*]| = 2$ and $G \cong S_2$ (like instances 1–2).

- For a 3-node undirected cycle $X—Y—Z—X$, the equivalence class contains all acyclic orientations: $||G^*|| = 3 \cdot 2 = 6$ DAGs (minus cyclic ones), with $G \cong S_3$ (symmetric group on 3 elements).
- For larger undirected components, $||G^*||$ can grow combinatorially, giving orbits of arbitrary size.

The unfaithfulness bound $U \geq 1/|G \cdot \theta|$ (from Theorem 55) therefore yields $U \geq 1/2$ for chains but $U \geq 1/6$ for triangles and potentially much smaller for larger structures.

Theorem 59 (Causal Discovery Impossibility). *For any CPDAG with $||G^*|| \geq 2$ DAGs in its Markov equivalence class, no edge orientation rule can be simultaneously faithful (matching the data-optimal DAG), stable (invariant to resampling), and complete (orienting all undirected edges).*

The achievable set has exactly two families:

- **Family $\mathcal{A}''$ (single-dataset):** Orient edges from one dataset. Faithful and complete but unstable. Unfaithfulness $U'' = (||G^*|| - 1)/||G^*||$ for within-class edge orientations.
- **Family $\mathcal{B}''$ (averaged / conservative):** Report undirected edges for ambiguous orientations (i.e., output the CPDAG). Stable with ties. Unfaithfulness $U'' = 0$.

Proof. We verify the premises of the Symmetric Bayes Dichotomy (Theorem 55).

G -invariance of π . For any two DAGs $G_1, G_2 \in [G^*]$, the Markov equivalence guarantees that they encode the same set of conditional independences. By the faithfulness assumption, the observational distribution is compatible with both G_1 and G_2 . Any statistical test based on finite data cannot distinguish between G_1 and G_2 at the population level (they imply identical distributions). Therefore π is G_{CPDAG} -invariant: the distribution over datasets is identical regardless of which member of $[G^*]$ generated the data.

Finite-sample noise reverses optimal choice. With finite data, empirical conditional independence tests have estimation error $O(1/\sqrt{n})$. For edges that are undirected in the CPDAG (i.e., the data cannot determine the orientation), different random samples will favor different orientations. This is the Rashomon property for edge orientations.

Application of Theorem 55. Part (i): any complete orientation rule (selecting one DAG from $[G^*]$) disagrees with $(||G^*|| - 1)/||G^*||$ of datasets. Part (ii): the Bayes estimator under G_{CPDAG} -invariance reports the CPDAG (ties for undirected edges). Part (iii): the ideal (stable, faithful, complete) is infeasible. Families $\mathcal{A}''$ and $\mathcal{B}''$ follow. $\square$

What this instance adds. Instances 1–2 (feature attribution and model selection) both have binary orbits ($|G \cdot \theta| = 2$) with unfaithfulness bound $U \geq 1/2$. Instance 3 has variable-size orbits ($|G \cdot \theta| = 2, 6, \dots$) with unfaithfulness bound $U \geq 1/||G^*||$, which can be much smaller than $1/2$ for large equivalence

classes. This demonstrates that the Symmetric Bayes Dichotomy is not an artifact of binary symmetry but applies to arbitrary finite group actions, with the unfaithfulness bound adapting to the orbit structure.

The practical implication: algorithms like PC and GES correctly output CPDAGs (Family $\mathcal{B}''$) precisely because complete orientation of Markov-equivalent edges is impossible from observational data alone. Interventional data breaks the symmetry (analogous to conditional SHAP breaking symmetry when $\beta_j \neq \beta_k$).

Corollary 60 (Causal discovery as SBD instance). *The Causal Discovery Impossibility (Theorem 59) is an instance of the Symmetric Bayes Dichotomy with $\Theta = \text{edge orientations in } [G^*]$, $G = G_{\text{CPDAG}}$ (automorphism group of the CPDAG), $\mathcal{D} = \text{observational datasets}$.*

27 LLM Attention Attribution Instability

The impossibility is not limited to tabular models. We demonstrate empirically that attention-based token importance in transformers exhibits the same instability.

Setup. We load DistilBERT (distilbert-base-uncased) and create 10 model variants by adding small random perturbations ($\sigma = 0.02$) to the attention weights, simulating different training runs. For 10 test sentences, we compute attention-based token importance (mean attention received from all heads and layers) and check whether the relative ranking of adjacent tokens (e.g., “very” vs. “good”) flips across the 10 variants.

Results. Of 59 adjacent token pairs across the test sentences, 88% have flip rates exceeding 10%. Selected examples:

Token A	Token B	Flip rate	Interpretation
“extremely”	“poor”	50%	Coin flip
“not”	“bad”	40%	Highly unstable
“absolutely”	“terrible”	40%	Highly unstable
“particularly”	“sharp”	10%	Borderline

The mean Spearman correlation of full-sentence token rankings across model variants is 0.637 — moderate instability, comparable to single-model SHAP instability on tabular data with moderate collinearity ($\rho \approx 0.5$).

Interpretation. Adjacent tokens in natural language carry correlated information (“very” and “good” jointly convey sentiment strength). This correlation induces the same Rashomon structure: different model variants assign different relative importance to the correlated tokens, exactly as the Symmetric Bayes Dichotomy predicts. Token-level explanations of LLMs (attention maps, gradient-based attribution) face the same impossibility as feature-level explanations of tabular models.

The resolution is analogous: average attention across multiple model variants to produce stable token importance estimates, accepting ties for tokens carrying correlated information.

27.1 Fine-tuning replication

The weight perturbation experiment above is a proxy for retraining variation. To verify the finding holds under actual training stochasticity, we fine-tune DistilBERT on SST-2 (2,000 training samples) with 5 different random seeds (1 epoch each, AdamW, $\text{lr}=2\times 10^{-5}$, batch size 16) and repeat the token importance analysis.

Results. Of 55 adjacent token pairs across 10 test sentences, 14.5% (8 pairs) have flip rates exceeding 10% across the 5 fine-tuned models. The mean Spearman correlation of full-sentence token rankings is 0.956 (minimum: 0.817).

Method	% Unstable ($> 10\%$ flip)	Mean Spearman
Weight perturbation ($\sigma=0.02$, 10 models)	88%	0.637
Fine-tuning (5 seeds, 1 epoch)	14.5%	0.956

The lower instability rate under fine-tuning is expected: 1-epoch fine-tuning produces less weight divergence than a 2% Gaussian perturbation of all parameters. However, instability is confirmed under realistic training variation: specific token pairs (e.g., “particularly” vs. “sharp”) reach flip rates of 60% across fine-tuned models. The perturbation result establishes an upper bound on Rashomon set diversity; the fine-tuning result confirms the effect persists under standard training. To test whether longer training increases instability, we repeat the experiment with 3 epochs. The instability rate is unchanged:

Method	Epochs	% Pairs unstable	Mean Spearman
Weight perturbation	—	88%	0.637
Fine-tuning	1	14.5%	0.956
Fine-tuning	3	14.5%	0.965

The identical 14.5% across 1 and 3 epochs indicates the instability is structural (driven by collinearity in the attention patterns), not a training-depth artifact. The same three token pairs remain unstable at both training lengths (“not/bad,” “particularly/sharp,” “surprisingly/low”). The gap between fine-tuning (14.5%) and perturbation (88%) reflects the difference in weight divergence: fine-tuning updates parameters by $O(0.001)$ via gradient descent, while perturbation adds $O(0.02)$ noise to all parameters. The perturbation result represents the full Rashomon set; the fine-tuning result represents the subset explored by standard training.

Scope and limitations. The attention-based instability reported here (14.5% of adjacent token pairs under fine-tuning) is preliminary evidence, not a formal extension of the impossibility theorem. Attention weights are not SHAP values: they do not satisfy the proportionality axiom, and the “collinear group” structure of tree features does not map cleanly to token positions. The result demonstrates that attribution instability under model multiplicity is not unique to tree ensembles, but a formal impossibility for transformer attention would require different axioms and definitions. We report this as suggestive evidence for the generality of the phenomenon, not as a proved extension.

28 Replication Study: Published SHAP Rankings Are Unstable

Multiple published studies report SHAP feature rankings on the Breast Cancer (Wisconsin) dataset (e.g., Lundberg & Lee, 2017). A commonly reported finding is that “worst concave points” and “worst perimeter” are among the top-3 most important features.

Using our 50-model validation (same dataset, same XGBoost configuration as standard practice: $T=100$, $\text{depth}=6$, $\eta=0.1$), we find that the ranking of “worst perimeter” vs. “worst area” ($|\rho| = 0.98$) **flips in 48% of training runs**. The ranking of “mean concave points” vs. “mean concavity” ($|\rho| = 0.96$) flips in 42% of runs.

Feature pair	$ \rho $	Flip rate	Published as stable?
worst perimeter $\leftrightarrow$ worst area	0.98	48%	Yes
mean concavity $\leftrightarrow$ mean concave pts	0.96	42%	Yes
worst radius $\leftrightarrow$ worst perimeter	0.99	36%	Yes
mean radius $\leftrightarrow$ mean perimeter	0.99	28%	Yes

Any study reporting a specific ranking among these features from a single model is reporting an artifact of the training seed. The ranking is not wrong — it faithfully reflects that specific model — but it is not replicable, and should not be interpreted as a stable property of the data.

This finding has implications for the **reproducibility** of scientific studies that use feature importance to identify “key predictors”: if the dataset has correlated features (68% of datasets do, per our prevalence survey), a single-model SHAP analysis may produce results that do not replicate.

28.1 The Ranking Lottery

We extend the replication study to 5 public datasets, counting how many distinct top- K SHAP rankings appear across 50 seeds (Table 12).

Seven distinct features compete for the top-3 positions on Breast Cancer: worst concave points, mean concave points, worst area, worst concavity, worst texture, worst perimeter, and area error. The #1 position alternates: worst

Table 12: The ranking lottery: distinct top-3 and top-5 SHAP rankings across 50 seeds (**subsample=0.8**, fixed test set). Deterministic control (**subsample=1.0**): 1 ranking for all datasets.

Dataset	Samples	Distinct top-3	Top-3 agree	Distinct top-5	Top-5 agree
Breast Cancer	569	24	4.2%	46	0.4%
Diabetes	768	2	88.5%	4	68.2%
Wine Quality	1,599	1	100%	2	55.6%
Heart Disease	270	1	100%	1	100%
California Housing	20,640	1	100%	1	100%

concave points (50%), mean concave points (32%), worst area (18%). At 100 seeds: 35 distinct top-3 (up from 24) and 84 distinct top-5 (up from 46)—the count keeps growing, confirming 24 is a lower bound on the number of possible rankings.

Subsample sensitivity. The instability persists even at reduced stochasticity. On Breast Cancer (50 seeds):

subsample	Distinct top-3	Agreement
0.80	24	4.2%
0.85	23	5.2%
0.90	22	6.9%
0.95	17	10.7%
1.00	1	100.0%

Only fully deterministic training eliminates the lottery. At **subsample=0.95**, 17 distinct rankings remain. “Just use less stochasticity” is not a resolution.

Cross-implementation. On Breast Cancer: XGBoost 24 distinct top-3 (4.2% agreement), LightGBM 29 (3.6%, **subsample_freq=1**), Random Forest 40 (2.5%, permutation importance). The instability is implementation-independent.

Gene count robustness. On AP_Colon_Kidney, the ranking lottery result is robust to the number of pre-selected genes: 8 distinct top-3 at 20 genes, 5 at 50, 7 at 100. The instability is not an artifact of the filtering threshold.

28.2 Gene Expression: Pathway-Level Consequence

On a 10,935-gene colon-vs-kidney expression dataset (AP_Colon_Kidney, 546 samples, top 50 genes by variance, OpenML), the #1 biomarker gene alternates between TSPAN8 (203824_at, invasion/metastasis pathway, 80% of seeds) and CEACAM5/CEA (201884_at, immune evasion pathway, 20% of seeds). Both are cell adhesion molecules enriched in colon tissue ($\rho = 0.858$), but with distinct cancer biology roles: TSPAN8 in invasion/metastasis signaling via

tetraspanin-enriched microdomains, and CEACAM5 (the classical clinical colon cancer marker CEA) in immune evasion and differentiation via GPI-anchored adhesion.

The broad pathway-level conclusion (colon identity genes drive the distinction) is preserved, but the specific biomarker target differs. A biomarker development pipeline reporting TSPAN8 as the top discriminator would target a different gene than one reporting CEA. DASH consensus ranks both in the top-2 (TSPAN8: importance 1.913; CEACAM5: 1.553), correctly reporting that both contribute without privileging either.

Probe-to-gene mappings verified via the Affymetrix HG-U133A annotation database.

Positive control. On AP_Breast_Lung (breast vs lung, 470 samples, 10,935 genes), the top-3 genes are stable (2 distinct rankings, 92% agreement) despite 171 highly correlated gene pairs ($|\rho| > 0.8$), because the top gene’s importance (2.90) dominates the second (1.40) with a clear gap.

29 Connection to Compressed Sensing

The attribution impossibility has a precise analogue in compressed sensing (CS). In CS, the *measurement coherence* $\mu(X) = \max_{j \neq k} |X_j^T X_k| / (\|X_j\| \|X_k\|)$ determines when sparse recovery is possible: unique recovery of an s -sparse signal requires $\mu < 1/(2s-1)$ (Donoho & Elad, 2003). When coherence is high (collinear measurements), the signal cannot be uniquely attributed to specific measurements.

In our setting, the correlation $|\rho_{jk}|$ plays the role of coherence, and the Z-test statistic Z_{jk} plays the role of the incoherence condition:

	Compressed sensing	Attribution stability
Measurements	Sensing matrix columns	Feature vectors
Coherence	$\mu = \max X_j^T X_k / (\ X_j\ \ X_k\)$	$ \rho_{jk} = \text{Corr}(X_j, X_k) $
Condition	$\mu < 1/(2s-1)$	$Z_{jk} > 1.96$
Success	Unique sparse recovery	Stable feature ranking
Failure	Non-unique recovery	Ranking flips (Rashomon)
Resolution	Basis pursuit (convex relaxation)	DASH (ensemble averaging)

The parallel is structural: both impossibilities arise because collinearity/coherence prevents unique identification of which input components (features/measurements) carry the signal. Both are resolved by relaxing uniqueness: CS uses convex relaxation (allowing fractional assignment), while DASH uses ensemble averaging (producing ties for indistinguishable features).

This connection suggests that information-theoretic lower bounds from CS (e.g., mutual information bounds on sparse recovery under coherence) could provide alternative proofs of the attribution impossibility and potentially tighter quantitative bounds.

30 Fairness Audit Impossibility

The attribution impossibility has a direct and previously unrecognized consequence for algorithmic fairness auditing.

Setup. Consider a model trained on features $X_1, \dots, X_P$ that include a protected proxy: feature j is correlated with a protected attribute A (e.g., zip code correlates with race, $|\rho_{jA}| > 0$). Feature j is also collinear with a non-protected feature k ($|\rho_{jk}| > \rho_0$, e.g., zip code and income). A fairness auditor uses SHAP to determine whether the model “relies on” the protected proxy by checking whether j ranks among the top- K important features.

Theorem 61 (Fairness Audit Impossibility). *If features j (protected proxy) and k (non-protected) satisfy:*

(F1) *Collinearity: $|\rho_{jk}| > \rho_0$ for threshold ρ_0 ,*

(F2) *Similar true importance: $|\mu_j - \mu_k| < \sigma_{jk} \cdot z_\alpha / \sqrt{M}$ (the attribution gap is smaller than the detection threshold),*

then for a single-model audit ($M=1$), the probability that the auditor concludes “the model relies on the protected proxy” (i.e., ranks j above k) is:

$$\Pr[\text{audit flags proxy}] = \frac{1}{2} \pm \varepsilon_{BE},$$

where $\varepsilon_{BE} \leq 0.4748\gamma_{jk}/\sigma_{jk}^3$ (Berry–Esseen). The audit conclusion is a coin flip.

More precisely: given two independent training runs producing models f, f' , the probability that the audit reaches opposite conclusions about proxy reliance is:

$$\Pr[\text{audit}(f) \neq \text{audit}(f')] = \frac{1}{2} \pm \varepsilon_{BE}.$$

Proof. Direct corollary of the Attribution Impossibility (Theorem 1) applied to the pair (j, k) .

By conditions (F1)–(F2), features j and k are in the same collinear group and have similar population-level attributions. The Rashomon property (Theorem S3, Rashomon inevitability) holds: $\Pr[\varphi_j(f) > \varphi_k(f)] = 1/2$ over training runs.

A single-model audit that ranks j above k when $\varphi_j(f) > \varphi_k(f)$ therefore concludes “proxy reliance” with probability $1/2$. The Berry–Esseen bound follows from Theorem F1.

For two independent audits: they agree iff both models rank j, k the same way, which occurs with probability $1/2$ (by independence and the $1/2$ marginal). $\square$

Implications for AI governance. This result means that a regulator conducting a SHAP-based proxy audit on a model with collinear features is, for affected feature pairs, making decisions with the statistical reliability of a coin flip. Two auditors examining the *same model architecture trained on the same data* with different random seeds may reach opposite conclusions about whether the model relies on a protected attribute.

Under the EU AI Act Art. 13(3)(b)(ii), providers must disclose “known and foreseeable circumstances” which may lead to risks to health, safety or fundamental rights. We argue that our theorem identifies such a circumstance: in our reading, SHAP-based proxy audits are unreliable for features that are collinear with both the protected attribute and a non-protected feature. The remedy is ensemble auditing: train M models and use DASH consensus. The required M follows from the ensemble lower bound (Proposition 31).

Scope. The impossibility applies when conditions (F1)–(F2) hold — i.e., when the protected proxy is genuinely interchangeable with a non-protected feature in terms of predictive value. When the protected feature has *unique* predictive power ($|\mu_j - \mu_k|$ is large), a single-model audit correctly identifies proxy reliance. The conditional SHAP threshold (Table 11) quantifies the boundary: at $\rho \geq 0.9$, the proxy must have nearly an order-of-magnitude stronger causal effect than the non-protected feature ($\Delta\beta^* \approx 0.9$) before the audit becomes reliable.

Example: lending. In credit models, “zip code” (a proxy for race) is often collinear with “income bracket” ($|\rho| \approx 0.6\text{--}0.8$). If both features have similar predictive power for default risk, a single-model SHAP audit that finds “zip code is the 3rd most important feature” may conclude proxy discrimination — but retraining with a different seed could produce a model where zip code drops to 8th and income rises to 3rd. The *audit conclusion about discrimination depends on the random seed*, not on the model’s actual reliance on the protected attribute.

Intersectional considerations. When multiple protected attributes (race, gender, age) are each proxied by collinear non-protected features, the impossibility applies independently to each proxy pair. The worst case is multiplicative: if K protected attributes each have unstable proxies, the probability that a single-model audit correctly identifies all K proxy reliance directions is $(1/2)^K$, exponentially vanishing in K . For $K = 3$ (a common intersectional setting), the joint correctness probability is $1/8 = 12.5\%$. A full analysis of intersectional instability under model multiplicity is left to future work.

Tightness transition under performance constraints. The framework’s tightness classification applies to the Chouldechova–Kleinberg fairness impossibility (calibration P_1 + equal FPR P_2 + equal FNR P_3 cannot coexist when base rates differ). Under performance constraints ($\text{TPR} \geq \tau$ for both groups):

Constraint	Tightness type	Achievable pairs
$\tau = 0$ (unconstrained)	Full	All three pairs (degenerate classifiers)
$\tau \geq 0.3$	PPV-blocked	Only P_2+P_3 (equalized odds)

At any practical performance level, calibration (PPV parity) is individually incompatible with both equal FPR (P_1+P_2 gap > 0.10) and equal FNR (P_1+P_3 gap > 0.13). The only achievable pairwise fairness compromise is equalized odds (P_2+P_3 gap < 0.006). Validated on Adult Income (sex, base rate gap 0.20) and synthetic data (base rate gap 0.25–0.40) with bootstrap model retraining (20 iterations, 100% stable). This finding changes the practical recommendation from “choose a fairness criterion” to “equalized odds is the only achievable criterion at practical performance; calibration is structurally incompatible.”

Reynolds best-approximation optimality. The orbit average $Rv = \frac{1}{|G|} \sum_{g \in G} g \cdot v$ is the *closest G -invariant vector* to v in L^2 norm. For any G -invariant w (i.e., any stable approximation): $\|v - Rv\| \leq \|v - w\|$. The proof is a direct consequence of the Pythagorean decomposition: $v = Rv + (v - Rv)$ where $Rv \in V^G$ and $(v - Rv) \perp V^G$. Lean: `reynolds_best_approximation` (idempotent self-adjoint projection). This establishes that DASH and V^G minimize information loss among *all* stable methods — not just unbiased or linear ones.

31 Expected Ranking Discordance

We derive a closed-form formula for the expected disagreement between two independent SHAP rankings, providing a predictive theory of SHAP replication failure.

Proposition 62 (Expected Kendall tau distance). *For a feature space with L collinear groups of sizes $m_1, \dots, m_L$ and between-group attribution gaps $\Delta_{\ell\ell'}$ with noise $\sigma_{\ell\ell'}$, the expected number of pairwise ranking disagreements (Kendall tau distance) between two independently trained models is:*

$$\mathbb{E}[\tau] = \underbrace{\sum_{\ell=1}^L \binom{m_\ell}{2} \cdot \frac{1}{2}}_{\text{within-group (coin flips)}} + \underbrace{\sum_{\ell < \ell'} m_\ell \cdot m_{\ell'} \cdot \Phi\left(-\frac{\Delta_{\ell\ell'}}{\sigma_{\ell\ell'}}\right)}_{\text{between-group (noise)}} \quad (9)$$

The within-group term is exact (from Proposition 13); the between-group term uses the Gaussian approximation (valid for continuous attribution distributions).

Proof. The Kendall tau distance counts the number of discordant pairs: $\tau = |\{(i, j) : i < j, \text{rank}_f(i) \neq \text{rank}_{f'}(i) \text{ relative to } j\}|$.

For a within-group pair (j, k) : by Proposition 13 Part (v), the flip rate is exactly $1/2$ (under tie-breaking). Each within-group pair contributes $1/2$ in expectation.

For a between-group pair (j, k) with j in group ℓ and k in group ℓ' : the flip rate is $\Phi(-\Delta_{jk}/\sigma_{jk})$ by Theorem 36 Part I. Since all $m_\ell \cdot m_{\ell'}$ cross-group pairs between groups ℓ and ℓ' share the same gap and noise (by within-group symmetry), each contributes equally. $\square$

Normalized discordance rate. Dividing by $\binom{P}{2}$ (total pairs) gives the expected *fraction* of rankings that fail to replicate:

$$\frac{\mathbb{E}[\tau]}{\binom{P}{2}} = \frac{\sum_{\ell} \binom{m_{\ell}}{2}/2 + \sum_{\ell < \ell'} m_{\ell} m_{\ell'} \Phi(-\Delta_{\ell\ell'}/\sigma_{\ell\ell'})}{P(P-1)/2}.$$

The first term is the *Rashomon coefficient* $R = \sum_{\ell} \binom{m_{\ell}}{2} / \binom{P}{2}$, which is computable from the correlation matrix alone (no model training required). The expected non-replication rate is $\geq R/2$ — a loose upper bound determined entirely by the feature correlation structure.

Validation on Breast Cancer. With 50 XGBoost models ($T=100$, depth=6, $\eta=0.1$, subsample=0.8):

Formula	Predicted $\mathbb{E}[\tau]$	Relative error
Naive (within-group = 1/2)	150.7	+312%
Refined (per-pair Φ)	26.8	-26.8%
Empirical mean Kendall tau distance		36.6

The naive formula severely overpredicts because it assumes all within-group pairs are coin flips. In reality, Breast Cancer’s 25-feature collinear group contains features with very different true importance (e.g., worst concave points vs. mean fractal dimension) — these are correlated but not symmetric, so their rankings are stable. The refined per-pair formula, which estimates $\Phi(-|\Delta_{ij}|/\sigma_{ij})$ for each pair from the model ensemble, reduces the error to 27%.

The Rashomon coefficient $R = 0.692$ predicts a non-replication rate lower bound of $R/2 = 34.6\%$; the actual rate is 8.4% — the bound is loose by $4\times$ because most within-group pairs have separable importance. The per-pair refinement is necessary for practical prediction; the Rashomon coefficient remains useful as a *worst-case* bound computable without training any models.

32 Attempted Common Generalization with Arrow’s Theorem

Arrow’s impossibility theorem for social choice and our Attribution Impossibility share a striking surface resemblance: both show that three desirable properties (faithfulness/Pareto, stability/IIA, completeness/non-dictatorship) cannot simultaneously hold. We investigated whether a common abstract framework subsumes both as special cases. The answer is *partially yes, but the unification is too shallow to yield new content*.

32.1 Structural comparison

The two theorems have fundamentally different mathematical structures:

	Arrow	Attribution Impossibility
Inputs	n voters' preferences $(L_1, \dots, L_n)$	Models $f \in \mathcal{F}$ (one at a time)
Output	Social ranking $f(L_1, \dots, L_n)$	Feature ranking σ
Aggregation	Many inputs $\rightarrow$ one output	One input $\rightarrow$ one output (per model)
Key axiom	IIA: $\sigma(a, b)$ depends only on voters' (a, b) rankings	Stability: σ is model-independent
Source	Preference profiles disagree	Rashomon property

The critical asymmetry. Arrow's theorem concerns a *profile aggregation* function $f: \mathcal{L}^n \rightarrow \mathcal{L}$ that takes all voters' preferences simultaneously. The impossibility arises because IIA, combined with Pareto and non-dictatorship, constrains this multi-input function too tightly.

The Attribution Impossibility concerns a *single-input* mapping: for each model f , the ranking should reflect f 's attributions. The impossibility arises because the ranking must simultaneously agree with all models but is supposed to be model-independent (stable). There is no profile aggregation—each model independently demands faithfulness.

32.2 Attempted common framework

Definition 63 (Abstract Aggregation Problem). *An abstract aggregation problem is a tuple $(\mathcal{I}, \mathcal{A}, \mathcal{R}, \text{Agree})$ where:*

- $\mathcal{I}$ is a set of instances (voters or models),
- $\mathcal{A}$ is the set of alternatives (candidates or features),
- $\mathcal{R}$ is the set of total orders on $\mathcal{A}$,
- $\text{Agree}: \mathcal{I} \rightarrow \mathcal{R}$ assigns each instance its “preferred” order.

An aggregation rule $F: 2^{\mathcal{I}} \rightarrow \mathcal{R}$ maps subsets of instances to a consensus order.

Arrow's setup: F takes the full profile $(L_1, \dots, L_n)$ and IIA says F 's pairwise comparison of (a, b) depends only on the restriction of each L_i to $\{a, b\}$.

Attribution setup: F is required to be a *constant function* (stability: the ranking does not depend on which model). Faithfulness says this constant ranking agrees with every instance's ordering.

The “common axiom” attempt. One might define *locality*: the consensus ranking of (a, b) depends only on instances' rankings of (a, b) . Arrow's IIA is locality restricted to the profile. Attribution stability is locality taken to its extreme: the ranking depends on *no* instance (it must be constant).

But this extreme makes the attribution version trivially impossible—a constant ranking cannot agree with conflicting instances (this is just the definition of the Rashomon property). Arrow’s theorem is much deeper: it shows that even when the aggregation *is* allowed to depend on instances, the constraints are still too tight. The proofs have entirely different structures.

32.3 Assessment

Remark 64 (No non-trivial common generalization). *The Attribution Impossibility and Arrow’s theorem are analogous but not instances of a common theorem in any non-trivial sense. One can define an abstract framework that subsumes both as shown above, but:*

1. *The common framework’s “impossibility theorem” would be: “if instances disagree and the aggregation rule must be constant, faithfulness is impossible.” This is a one-line observation, not a theorem.*
2. *Arrow’s actual content (the impossibility persists even when the rule is allowed to depend on all voters) has no analogue in the attribution setting.*
3. *The attribution impossibility’s actual content (the quantitative divergence $1/(1 - \rho^2)$, the resolution via ensemble averaging, the connection to the Rashomon set’s geometry) has no analogue in Arrow’s setting.*

The closest true statement is: both theorems arise from the tension between local coherence (respecting individual views) and global consistency (producing a single ranking), when the individuals disagree. This is a useful heuristic but not a formal unification.

Relationship to Sen’s liberalism paradox. A closer structural analogue may be Sen’s (1970) impossibility of a Paretian liberal, which also concerns a setting where individual “rights” (each person’s ranking of certain pairs should be respected) conflict with the Pareto criterion. The attribution impossibility can be viewed as a Sen-type result where each model has the “right” to determine the ranking of features (faithfulness), but these rights conflict across models. However, we do not pursue this further, as the connection remains at the level of analogy rather than formal subsumption.

33 Causal Identification Barrier Under Collinearity

We show that the attribution instability under collinearity has a direct causal counterpart: distinguishing the causal effects of correlated features requires interventional sample complexity that diverges as $\Omega(1/(1 - \rho^2))$ —the *same* rate as the attribution ratio.

33.1 Setup

Consider two features X_j, X_k with $\text{Corr}(X_j, X_k) = \rho$, jointly Gaussian with unit variances. The structural equation is $Y = \beta_j X_j + \beta_k X_k + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$.

The identification problem. From observational data, one can estimate $\hat{\beta}_j$ and $\hat{\beta}_k$ via OLS. But when $\rho \rightarrow 1$, the OLS estimator variance diverges: $\text{Var}(\hat{\beta}_j) = \sigma^2 / (n(1 - \rho^2))$. This is multicollinearity in its standard form.

Interventional experiments. Under the do-calculus, $\text{do}(X_j = x)$ breaks the $X_j \rightarrow X_k$ correlation. The interventional distribution is:

$$\mathbb{E}[Y \mid \text{do}(X_j = x)] = \beta_j x + \beta_k \mu_k$$

where $\mu_k = \mathbb{E}[X_k]$ under the intervened distribution. From a single-variable intervention on X_j , one can identify β_j cleanly—if the intervention fully breaks the correlation.

33.2 The barrier: soft interventions

In practice, interventions on collinear features are often *soft*: setting $X_j = x$ does not fully break the correlation. Under a soft intervention with residual correlation ρ_{post} :

$$X_k \mid \text{do}(X_j = x) \sim \mathcal{N}(\rho_{\text{post}} \cdot x, 1 - \rho_{\text{post}}^2).$$

The observed effect of the intervention is:

$$\mathbb{E}[Y \mid \text{do}(X_j = x)] = (\beta_j + \rho_{\text{post}} \beta_k) x + \text{const.}$$

When $\beta_j = \beta_k = \beta$ and $\rho_{\text{post}} \approx \rho$:

$$\mathbb{E}[Y \mid \text{do}(X_j = x)] \approx \beta(1 + \rho)x = \mathbb{E}[Y \mid \text{do}(X_k = x)].$$

The two interventional distributions become indistinguishable.

33.3 Sample complexity

Proposition 65 (Interventional sample complexity lower bound). *Consider testing $H_0 : \beta_j = \beta + \delta, \beta_k = \beta - \delta$ against $H_1 : \beta_j = \beta - \delta, \beta_k = \beta + \delta$ for some $\delta > 0$, using $\text{do}(X_j = x)$ interventions with residual correlation ρ_{post} .*

The number of interventional samples needed for a test with power $1 - \eta$ at significance α satisfies:

$$n_{\text{int}} \geq \frac{(z_\alpha + z_\eta)^2 \sigma^2}{4\delta^2(1 - \rho_{\text{post}})^2}.$$

When $\rho_{\text{post}} = \rho$ (the intervention does not reduce the correlation), this gives $n_{\text{int}} = \Omega(1/(1 - \rho)^2)$.

Proof. Under H_0 , the expected response to $\text{do}(X_j = x)$ is $(\beta + \delta + \rho_{\text{post}}(\beta - \delta))x$, while under H_1 it is $(\beta - \delta + \rho_{\text{post}}(\beta + \delta))x$. The difference in slopes is $2\delta(1 - \rho_{\text{post}})$. With noise variance σ^2 and n observations of the slope, the standard Neyman–Pearson sample size formula gives the result. $\square$

Remark 66 (Comparison with the attribution ratio). *The attribution ratio is $1/(1 - \rho^2) = 1/((1 - \rho)(1 + \rho))$. The causal sample complexity scales as $1/(1 - \rho_{\text{post}})^2$. These have the same leading singularity as $\rho \rightarrow 1$ (both diverge as $\Omega(1/(1 - \rho)^2)$ when $\rho_{\text{post}} = \rho$), but they are **not identical**:*

- *The attribution ratio is $1/(1 - \rho^2)$, which diverges as $1/(2(1 - \rho))$ near $\rho = 1$.*
- *The causal sample complexity is $\Omega(1/(1 - \rho)^2)$ — a faster divergence.*

This difference is meaningful: causal identification under soft interventions is harder than observational attribution instability. The attribution ratio captures the variance amplification (a ratio of quantities), while the causal barrier captures signal-to-noise (which squares the denominator).

The match is at the level of the singularity structure (both are controlled by $1 - \rho$) but not at the level of exponents. The shared singularity at $\rho = 1$ reflects a common root cause: collinearity makes features observationally interchangeable, whether the task is attribution or causal identification.

Hard interventions resolve both problems. When the intervention fully breaks the correlation ($\rho_{\text{post}} = 0$), the sample complexity reduces to $n_{\text{int}} = O(\sigma^2/\delta^2)$ —independent of ρ . This parallels the attribution impossibility: perfect interventions break the symmetry (just as knowing the true causal graph breaks the Rashomon property), while soft interventions preserve it.

34 Topological Analysis of the Impossibility

We investigate whether the Attribution Impossibility has a topological nature beyond simple connectedness arguments. The conclusion: for $m = 2$ and $m = 3$ features in a group, the topology adds **no content** beyond the intermediate value theorem. For $m \geq 4$, there is a mildly interesting observation about the permutohedron, but it does not yield quantitative improvements over the direct algebraic argument.

34.1 The $m = 2$ case: connectedness only

For two features j, k in a group, the attribution map assigns to each training seed $s \in \mathcal{S}$ a sign $\sigma(s) = \text{sign}(\varphi_j(f_s) - \varphi_k(f_s)) \in \{+1, -1\}$. The Rashomon property says both values are achieved.

A “stable ranking” is a constant function $\sigma^*: \mathcal{S} \rightarrow \{+1, -1\}$. The impossibility: σ^* cannot agree with $\sigma(s)$ for all s . This is immediate from the assumption that both signs are achieved; no topological machinery is needed.

If we additionally assume $\mathcal{S}$ is connected and σ is continuous (both strong assumptions), then σ must be constant—contradicting the Rashomon property. But the “continuous” assumption is unjustified for discrete training algorithms, and the result is weaker than what we already have (we prove impossibility without any continuity assumption).

34.2 The $m = 3$ case: still just connectedness

For three features, the attribution map sends seeds to rankings in S_3 (6 elements). Again, if $\mathcal{S}$ is connected and the map is continuous, then the image must be a connected subset of S_3 . Since S_3 is discrete, the image is a single point—contradicting Rashomon. This is the same connectedness argument; the symmetric group’s structure plays no role.

34.3 The $m \geq 4$ case: the permutohedron

For m features, the attribution vector $(\varphi_1(f_s), \dots, \varphi_m(f_s))$ lives in $\mathbb{R}^m$. The ranking is determined by the *chamber* of the *braid arrangement* (the hyperplane arrangement $\{x_i = x_j : i \neq j\}$) containing the attribution vector. The chambers are the cones of the *permutohedron*.

Remark 67 (No winding number content). *For $m \geq 3$, one might hope that the attribution map $s \mapsto (\varphi_1(f_s), \dots, \varphi_m(f_s))$ has a non-trivial winding number around the intersection of hyperplanes $\{x_j = x_k\}$, which would give a quantitative lower bound on the number of chamber crossings (ranking flips).*

This does not work for two reasons:

1. *The complement of the braid arrangement in $\mathbb{R}^m$ is simply connected for $m \geq 3$ (the codimension of each hyperplane is 1, so removing hyperplanes from $\mathbb{R}^m$ gives a space whose fundamental group is the pure braid group on m strands; but we would need the image to be a loop in this space, which requires additional structure not present in the attribution problem).*
2. *Even if a winding number existed, it would count signed crossings. The flip rate (unsigned crossings) is already captured by the direct algebraic bound $1/(1 - \rho^2)$ without topological machinery.*

Remark 68 (What topology could contribute). *There is one setting where topology adds genuine content: if the model space $\mathcal{S}$ has the structure of a manifold (e.g., a torus of hyperparameters) and the attribution map is smooth, then Morse theory could bound the number of critical points (ranking transitions). Specifically, if the attribution difference $g(s) = \varphi_j(f_s) - \varphi_k(f_s)$ is a Morse function on $\mathcal{S}$, the number of sign changes of g is bounded below by the sum of Betti numbers of $\mathcal{S}$.*

However, this requires: (a) $\mathcal{S}$ to be a smooth manifold, (b) g to be a Morse function, and (c) knowing the topology of $\mathcal{S}$. None of these hold for practical training algorithms (XGBoost’s seed space is discrete; neural net loss landscapes are non-smooth). We therefore do not pursue this direction.

34.4 Assessment

Remark 69 (Topology adds no content). *The Attribution Impossibility is fundamentally a combinatorial result (conflicting preferences cannot all be respected), not a topological one. The impossibility holds for arbitrary model spaces (including finite, discrete spaces with no topology). Attempts to extract topological content either reduce to the trivial connectedness argument or require unjustified smoothness assumptions that add no quantitative power beyond what the algebraic $1/(1 - \rho^2)$ ratio already provides.*

35 On Categorical/Axiomatic Strengthening

We investigate whether the impossibility can be strengthened to a metatheorem: “no finite axiom system on attributions can simultaneously guarantee faithfulness and stability under Rashomon.” The basic version is logically trivial; we identify a non-trivial quantitative extension.

35.1 The trivial version

Remark 70 (Trivial metatheorem). *Let $A_1, \dots, A_n$ be any finite set of axioms on an attribution method φ . If $A_1 \wedge \dots \wedge A_n$ implies faithfulness (the ranking reflects model attributions), stability (the ranking is model-independent), and completeness (the ranking decides all pairs), then $A_1 \wedge \dots \wedge A_n \rightarrow \perp$ under the Rashomon property.*

This is immediate: if the conjunction implies $F \wedge S \wedge C$, and $F \wedge S \wedge C \rightarrow \perp$ (Theorem 1), then the conjunction implies $\perp$. No content is added beyond Theorem 1.

35.2 The “constant attribution” escape

Why is a categorical strengthening difficult? Because trivial attribution methods escape the impossibility:

- $\varphi_j(f) = 1/P$ for all j, f (uniform): perfectly stable, perfectly equitable, satisfies any axiom system that doesn’t require faithfulness.
- $\varphi_j(f) = \mathbb{E}_f[\varphi_j(f)]$ (population mean): stable by construction, equitable under DGP symmetry.

Any “impossibility for all axiom systems” must exclude these methods, which means it must require faithfulness (or something implying it). But once faithfulness is required, we are back to Theorem 1.

35.3 A non-trivial quantitative version

The one direction that yields genuine content is a *quantitative* impossibility that does not require exact faithfulness.

Definition 71 (α -faithfulness). *An attribution method is α -faithful if for all models f and same-group features j, k :*

$$\Pr_f[\text{sign}(\varphi_j(f) - \varphi_k(f)) = \text{sign}(\sigma(j) - \sigma(k))] \geq \alpha$$

where σ is the stable ranking and the probability is over the training randomness.

Proposition 72 (Approximate faithfulness–stability tradeoff). *Under the Rashomon property with symmetric DGP (each model ranks $j > k$ or $k > j$ with equal probability $1/2$), any α -faithful stable ranking satisfies $\alpha \leq 1/2$.*

Proof. By DGP symmetry, for any same-group features j, k : $\Pr[\varphi_j(f) > \varphi_k(f)] = 1/2$. A stable ranking fixes $\sigma(j) > \sigma(k)$ or $\sigma(k) > \sigma(j)$. WLOG, suppose the stable ranking has $\sigma(j) > \sigma(k)$. Then α -faithfulness requires:

$$\Pr[\varphi_j(f) > \varphi_k(f)] \geq \alpha.$$

But $\Pr[\varphi_j(f) > \varphi_k(f)] = 1/2$ by symmetry, so $\alpha \leq 1/2$. $\square$

Remark 73. *This says: under perfect symmetry, no stable ranking can agree with the model’s attributions more than half the time. A coin flip achieves $\alpha = 1/2$. The stable ranking is no better than random for symmetric feature pairs.*

This result is non-trivial in that it holds for any attribution method satisfying the DGP symmetry condition, not just for specific axiom systems. However, it is a direct consequence of the symmetry—not of any deep axiomatic or categorical structure.

35.4 Beyond binary: the Spearman version

For m features in a group under full DGP symmetry, the attribution ranking is a uniform random permutation. A stable ranking $\sigma^* \in S_m$ has expected Spearman correlation with a uniform random permutation of:

$$\mathbb{E}[\rho_S(\sigma^*, \pi)] = 0$$

for any fixed σ^* , since $\mathbb{E}[\sum d_i^2]$ under a uniform random permutation equals $m(m^2 - 1)/3$, giving $\rho_S = 1 - 6 \cdot m(m^2 - 1)/(3 \cdot m(m^2 - 1)) = -1/(m - 1)$ for $m \geq 2$ (the expectation is actually slightly negative, not zero, by the classical result for random permutations).

Proposition 74 (Spearman bound for stable rankings). *Under full DGP symmetry within a group of m features, any stable ranking σ^* has expected Spearman correlation with the model’s attribution ranking satisfying:*

$$\mathbb{E}[\rho_S(\sigma^*, \pi)] = \frac{-1}{m - 1}.$$

In particular, the stable ranking is negatively correlated with the model’s ranking in expectation (for $m \geq 3$), and converges to zero correlation as $m \rightarrow \infty$.

Proof. Fix any $\sigma^* \in S_m$. By DGP symmetry, the model’s ranking π is uniform on S_m . The expected Spearman correlation of a fixed permutation with a uniform random permutation is $-1/(m - 1)$ (this is the classical result; see e.g., Kendall & Gibbons, 1990, Ch. 3). $\square$

35.5 Assessment

Remark 75 (Limited categorical content). *The “categorical impossibility” does not genuinely extend Theorem 1. The basic version (no finite axiom system implying $F \wedge S \wedge C$ can be consistent under Rashomon) is logically trivial. The quantitative version (Propositions 72 and 74) provides useful bounds but follows directly from DGP symmetry rather than from any categorical or axiomatic structure.*

The honest summary: the impossibility is a concrete result about conflicting requirements, not an instance of a general metatheoretic pattern. Attempts to “lift” it to a categorical level either (a) add no content, or (b) reduce to direct probabilistic calculations that do not require categorical language.

36 SymPy Verification

All algebraic consequences have been independently verified:

```
# dash-shap/paper/proofs/verify_lemma6_algebra.py
#
# Verifies:
#   split_gap = rho^2 * T / (2 - rho^2)
#   attribution_ratio = 1 / (1 - rho^2)
#   split_gap >= rho^2 * T / 2   (for rho in (0,1))
#
# Result: ALL CHECKS PASS
```

References

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